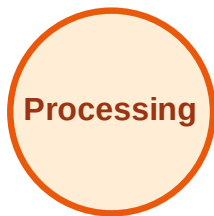


BFS Traversal

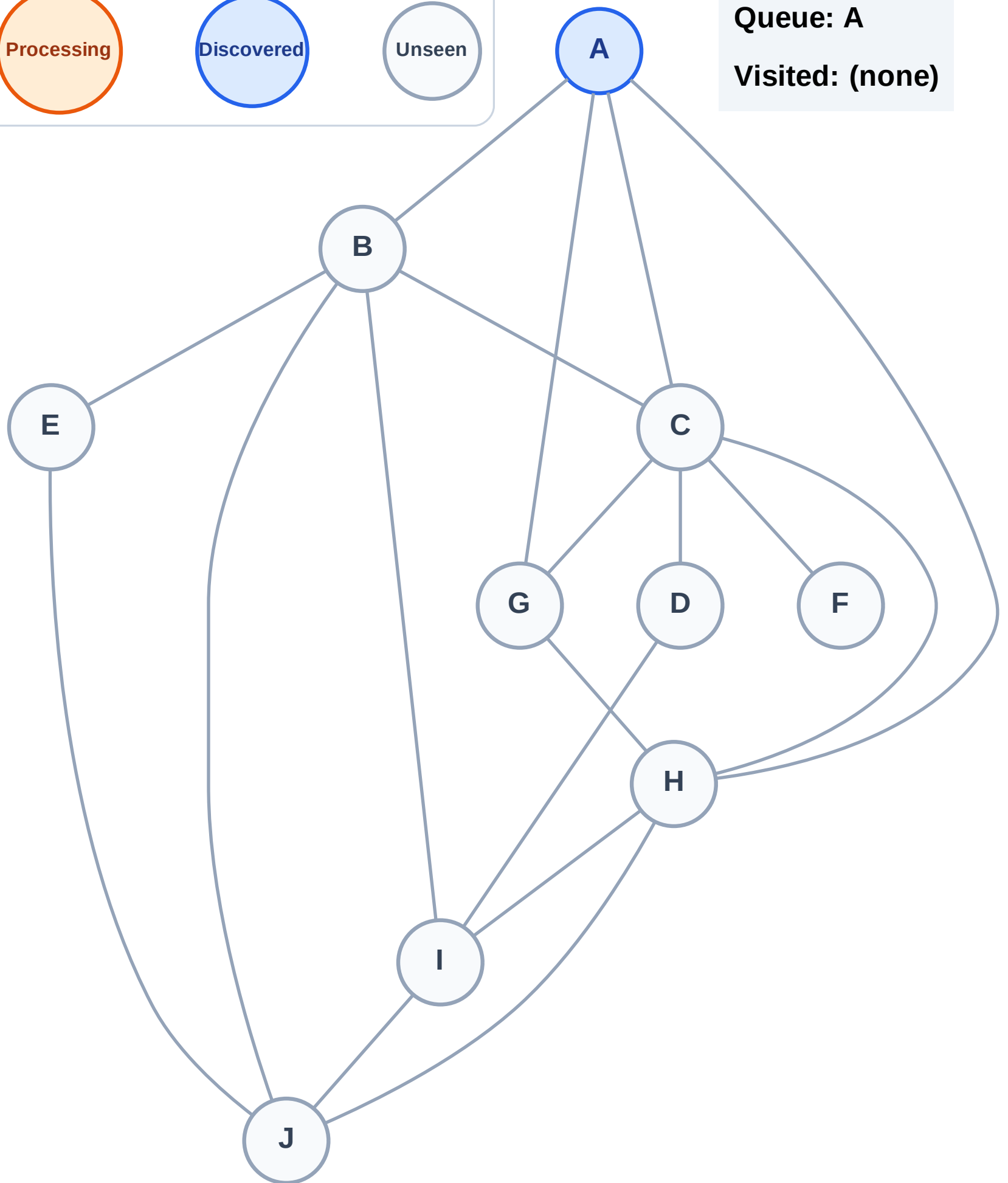
Step 1: Start at A

Legend



Queue: A

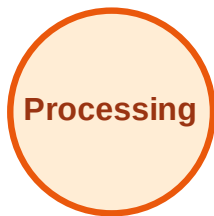
Visited: (none)



BFS Traversal

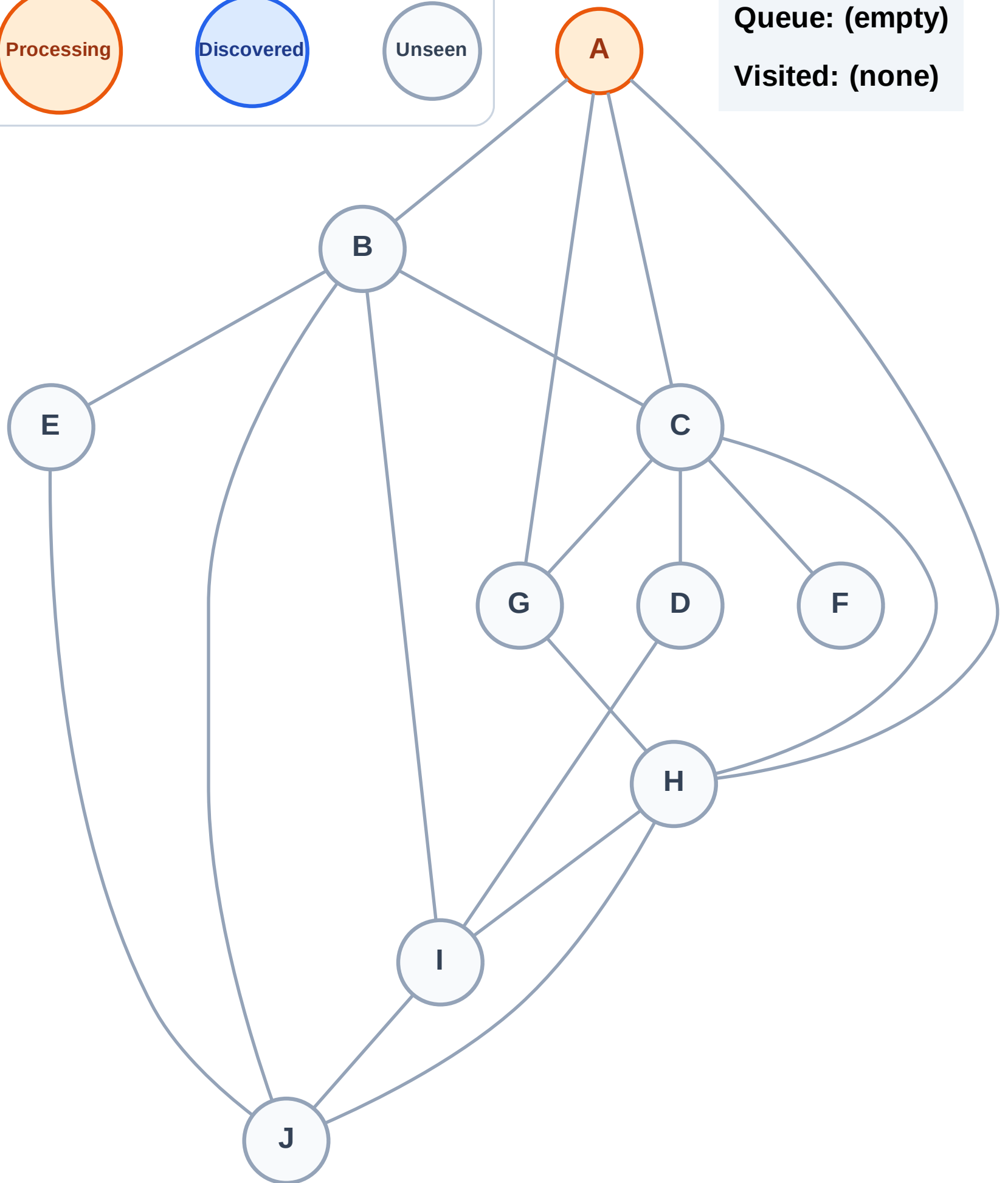
Step 2: Process node A

Legend



Queue: (empty)

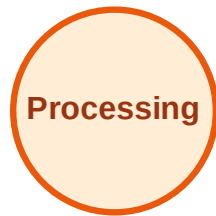
Visited: (none)



BFS Traversal

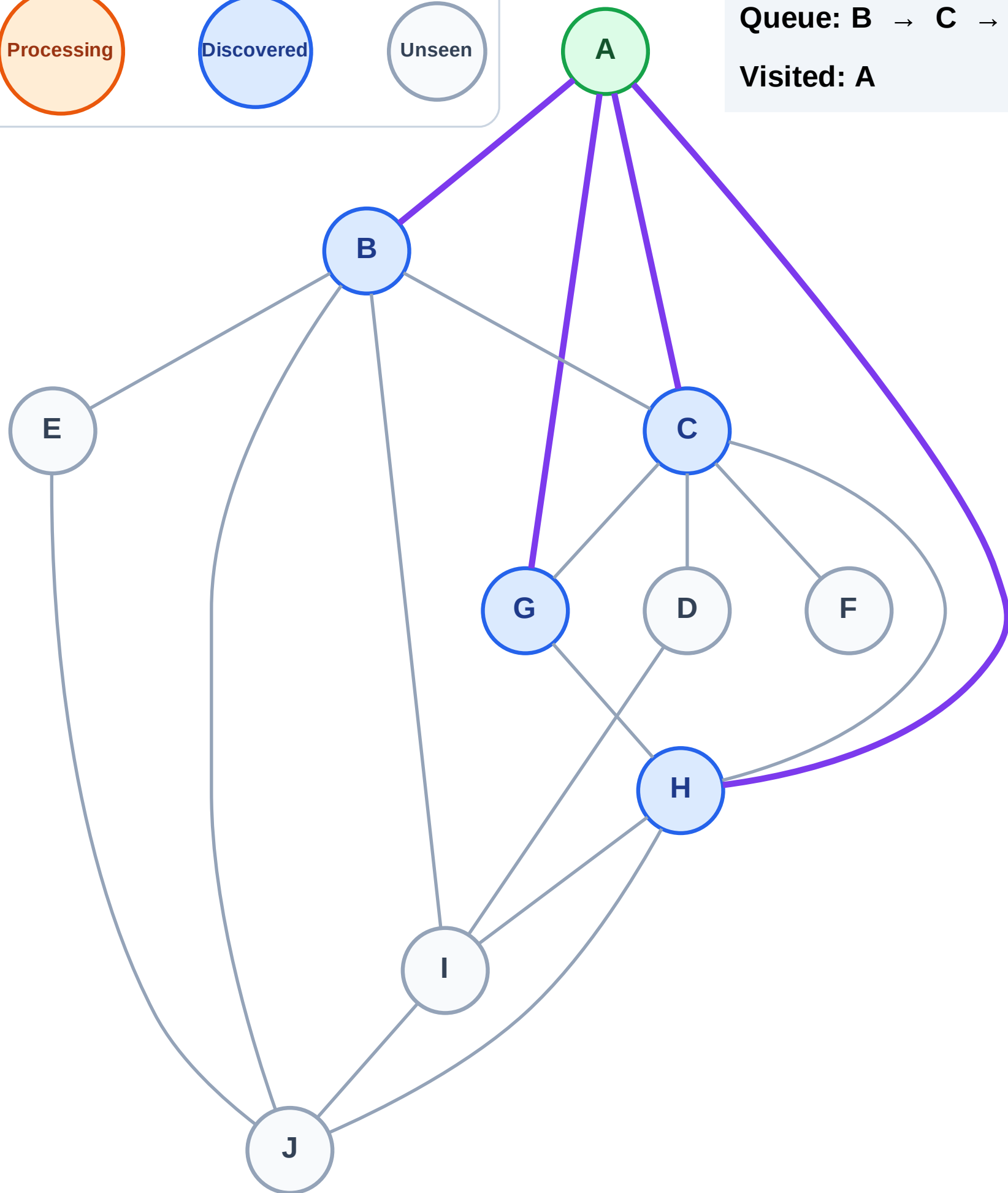
Step 3: Discover B, C, G, H from A

Legend



Queue: B → C → G → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

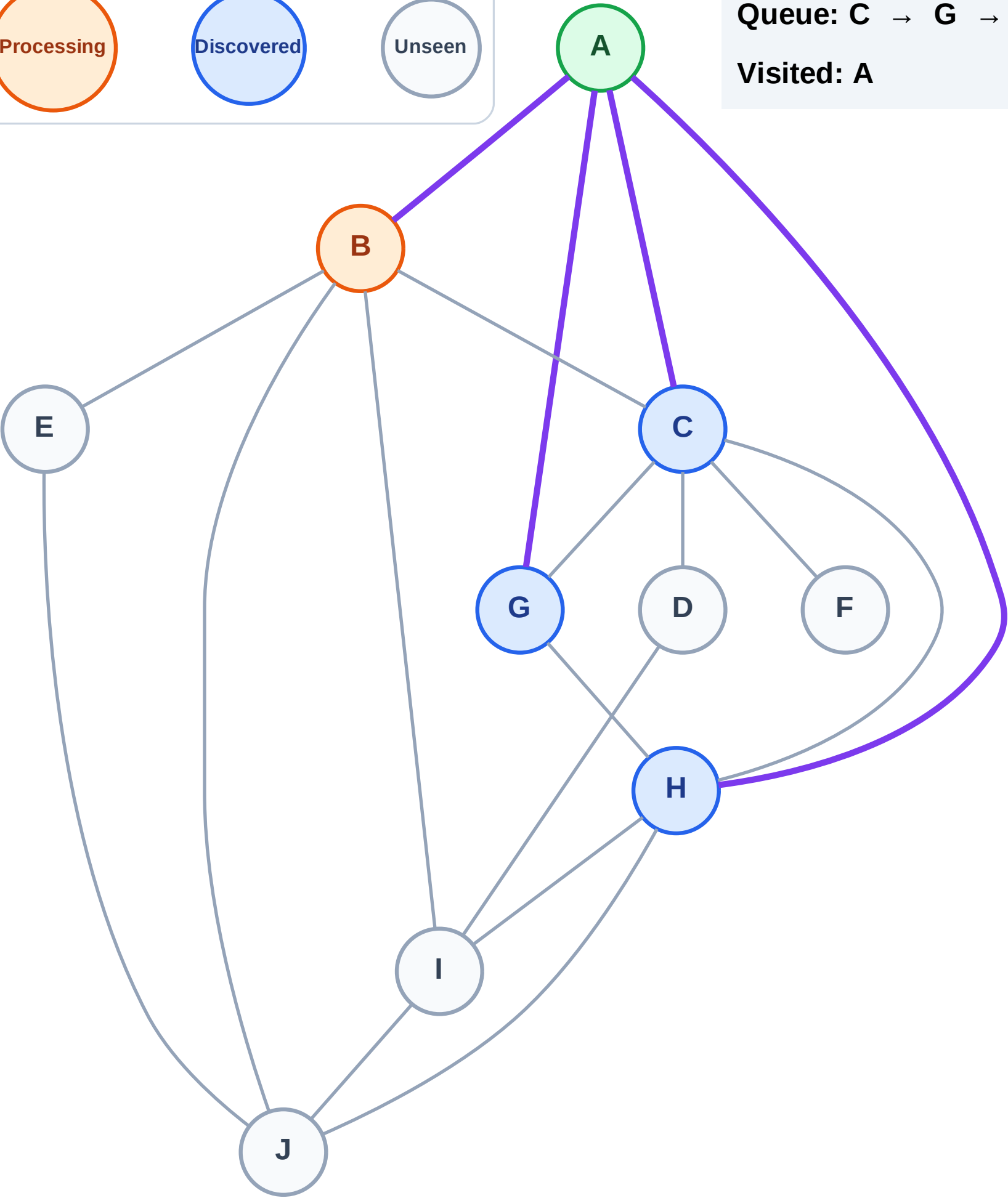
Complete

Processing

Discovered

Unseen

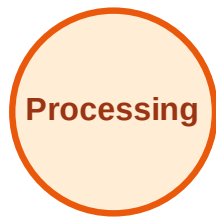
Queue: C → G → H
Visited: A



BFS Traversal

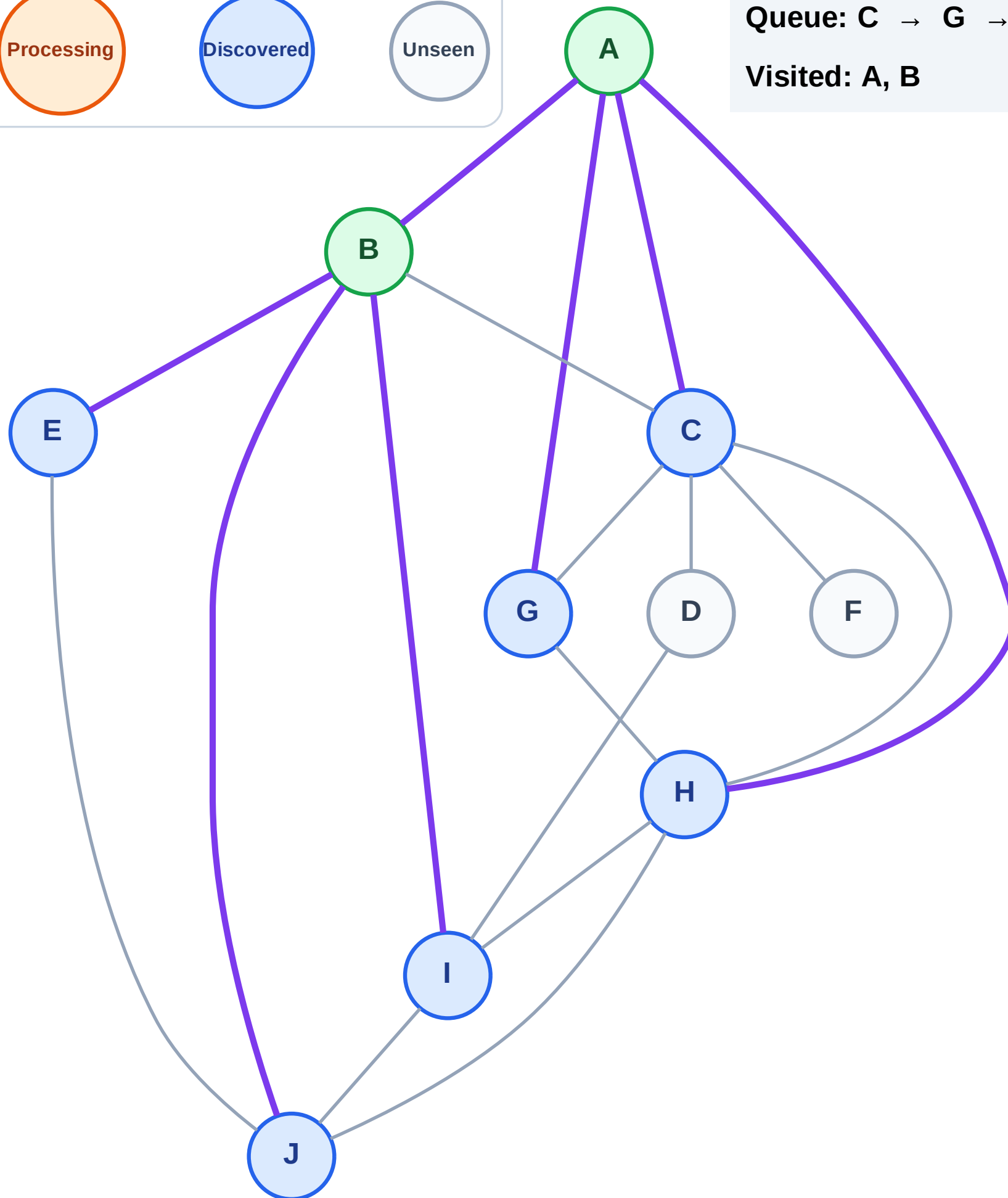
Step 5: Discover E, I, J from B

Legend



Queue: C → G → H → E → I → J

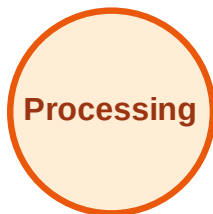
Visited: A, B



BFS Traversal

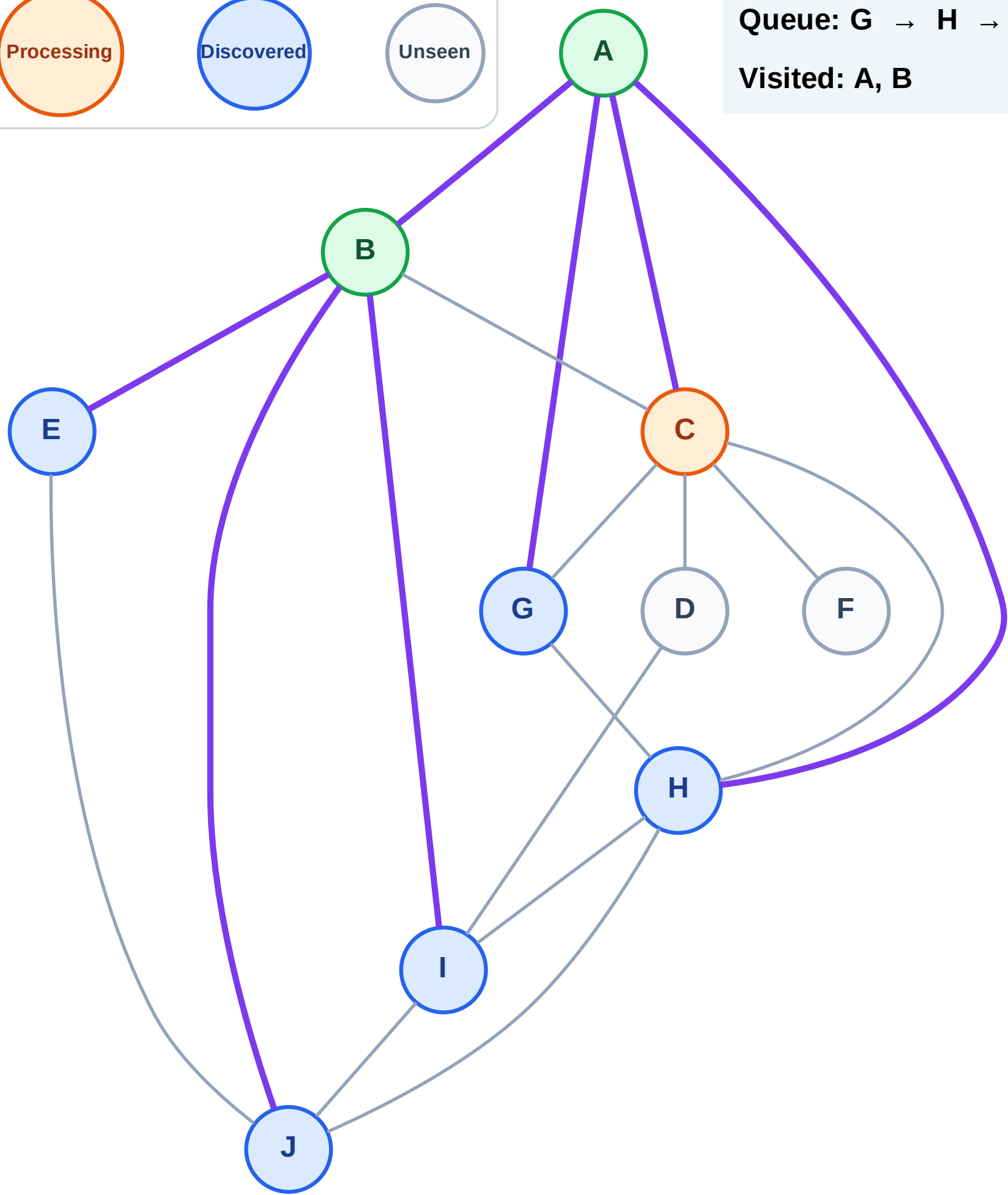
Step 6: Process node C

Legend



Queue: G → H → E → I → J

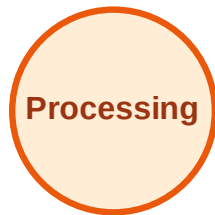
Visited: A, B



BFS Traversal

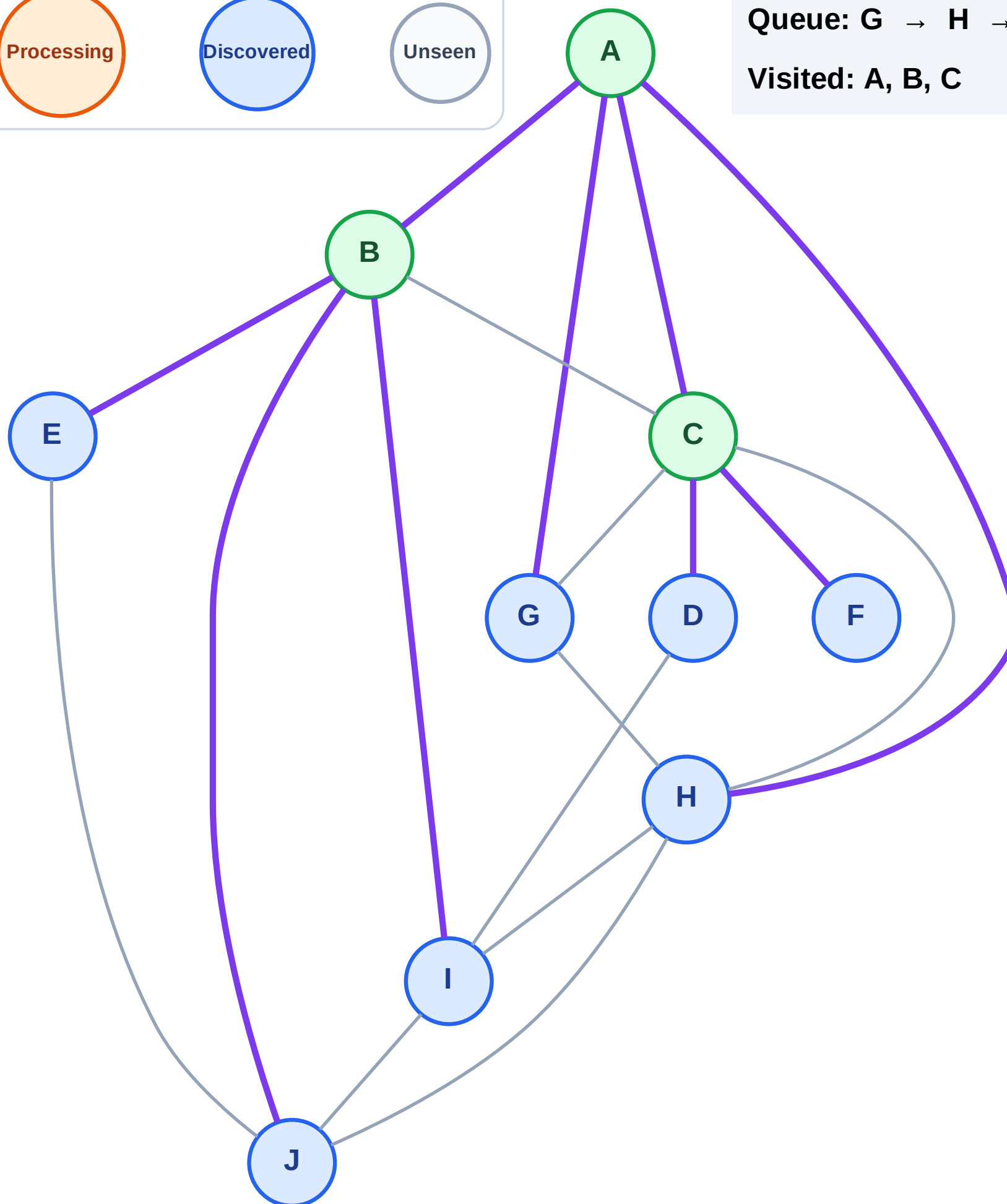
Step 7: Discover D, F from C

Legend



Queue: G → H → E → I → J → D → F

Visited: A, B, C

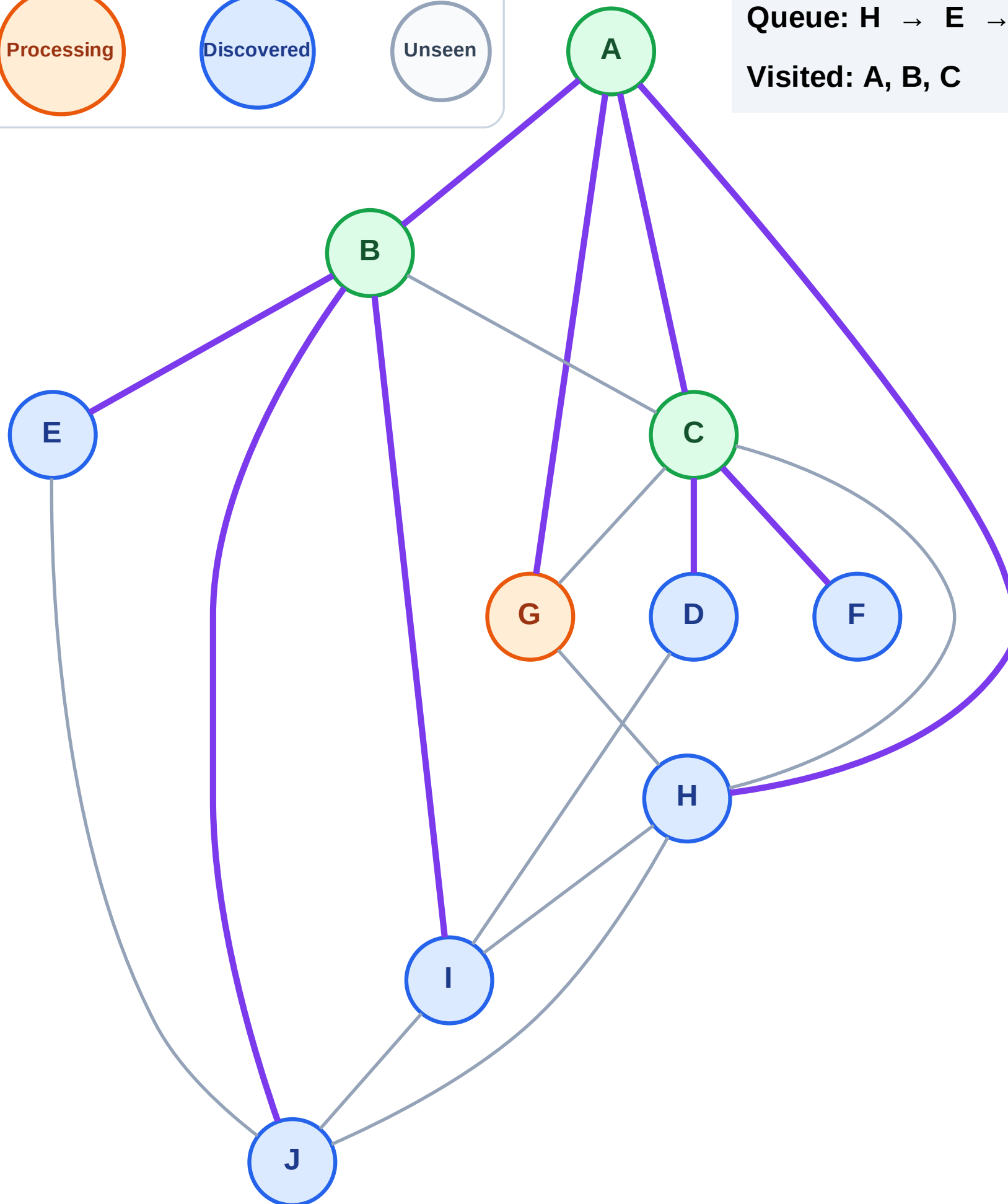
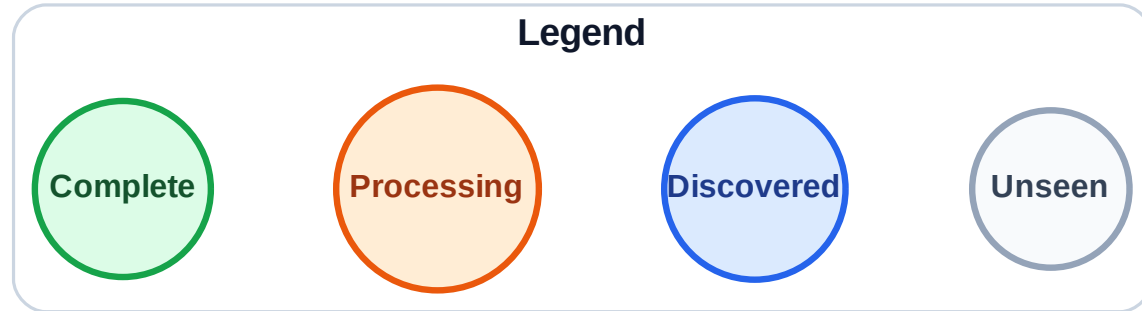


BFS Traversal

Step 8: Process node G

Queue: H → E → I → J → D → F

Visited: A, B, C



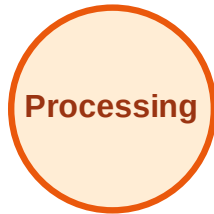
BFS Traversal

Step 9: No new neighbors from G

Legend



Complete



Processing



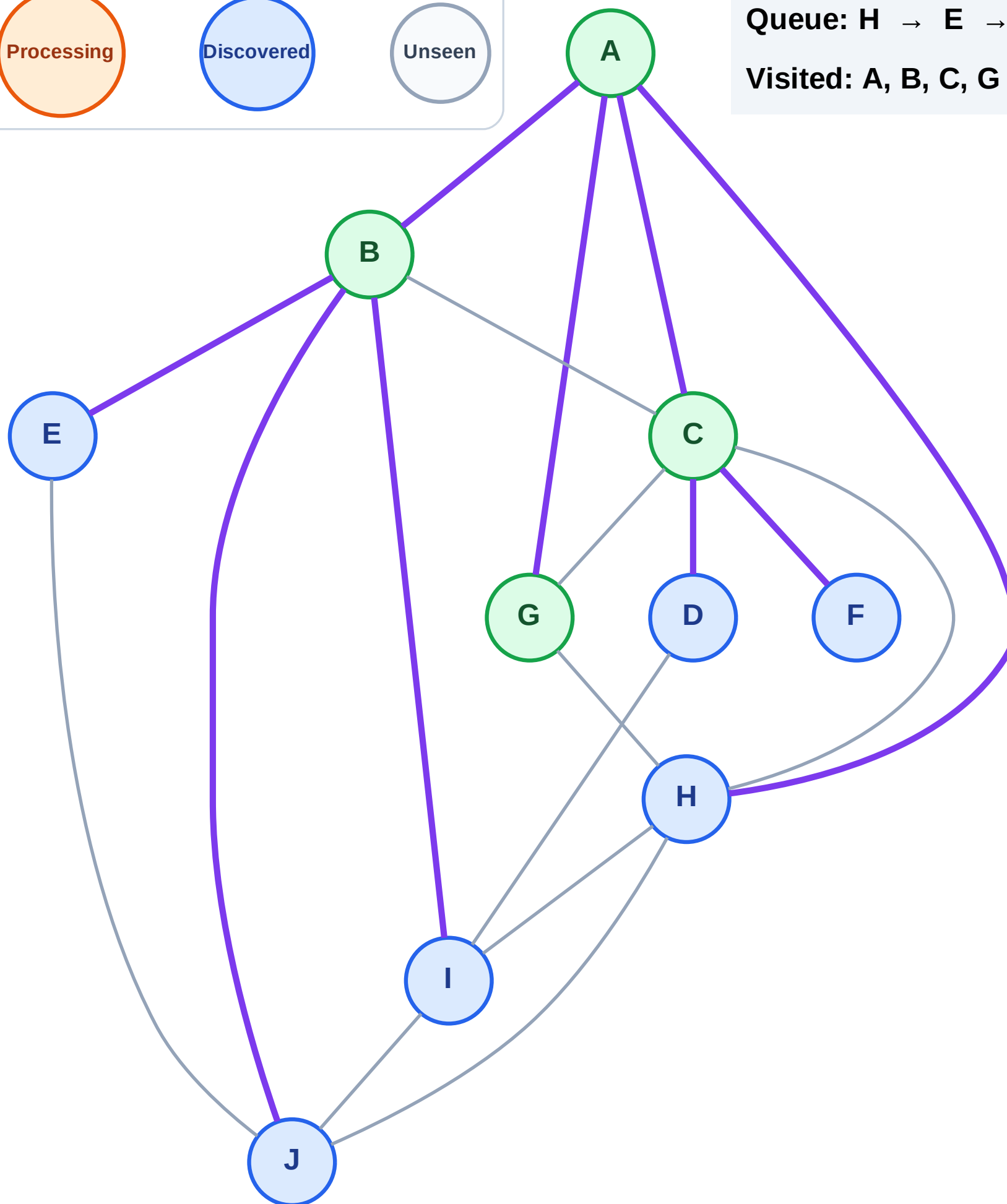
Discovered



Unseen

Queue: H → E → I → J → D → F

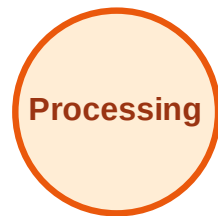
Visited: A, B, C, G



BFS Traversal

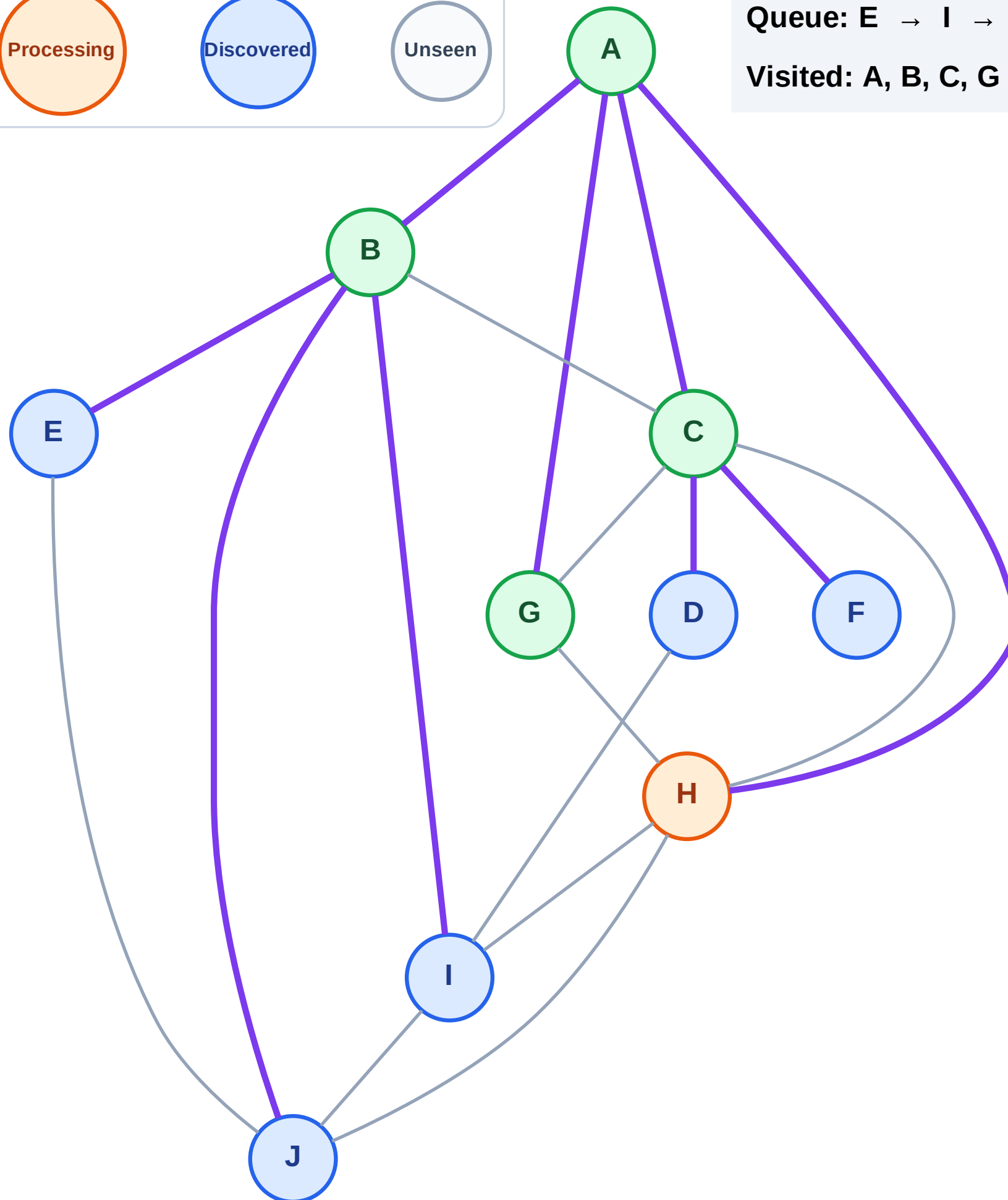
Step 10: Process node H

Legend



Queue: E → I → J → D → F

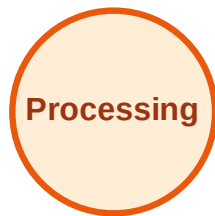
Visited: A, B, C, G



BFS Traversal

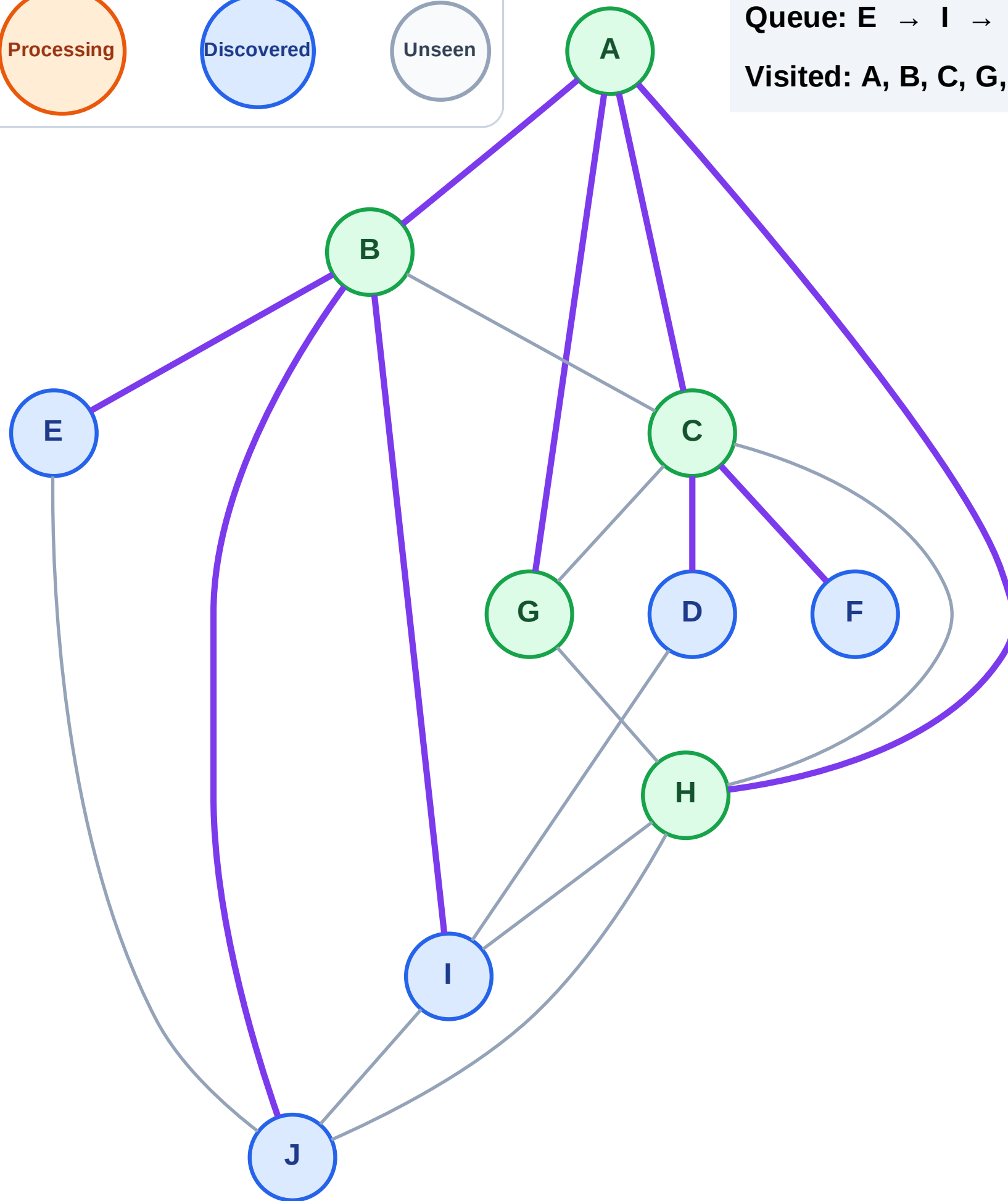
Step 11: No new neighbors from H

Legend



Queue: E → I → J → D → F

Visited: A, B, C, G, H



BFS Traversal

Step 12: Process node E

Legend

Complete

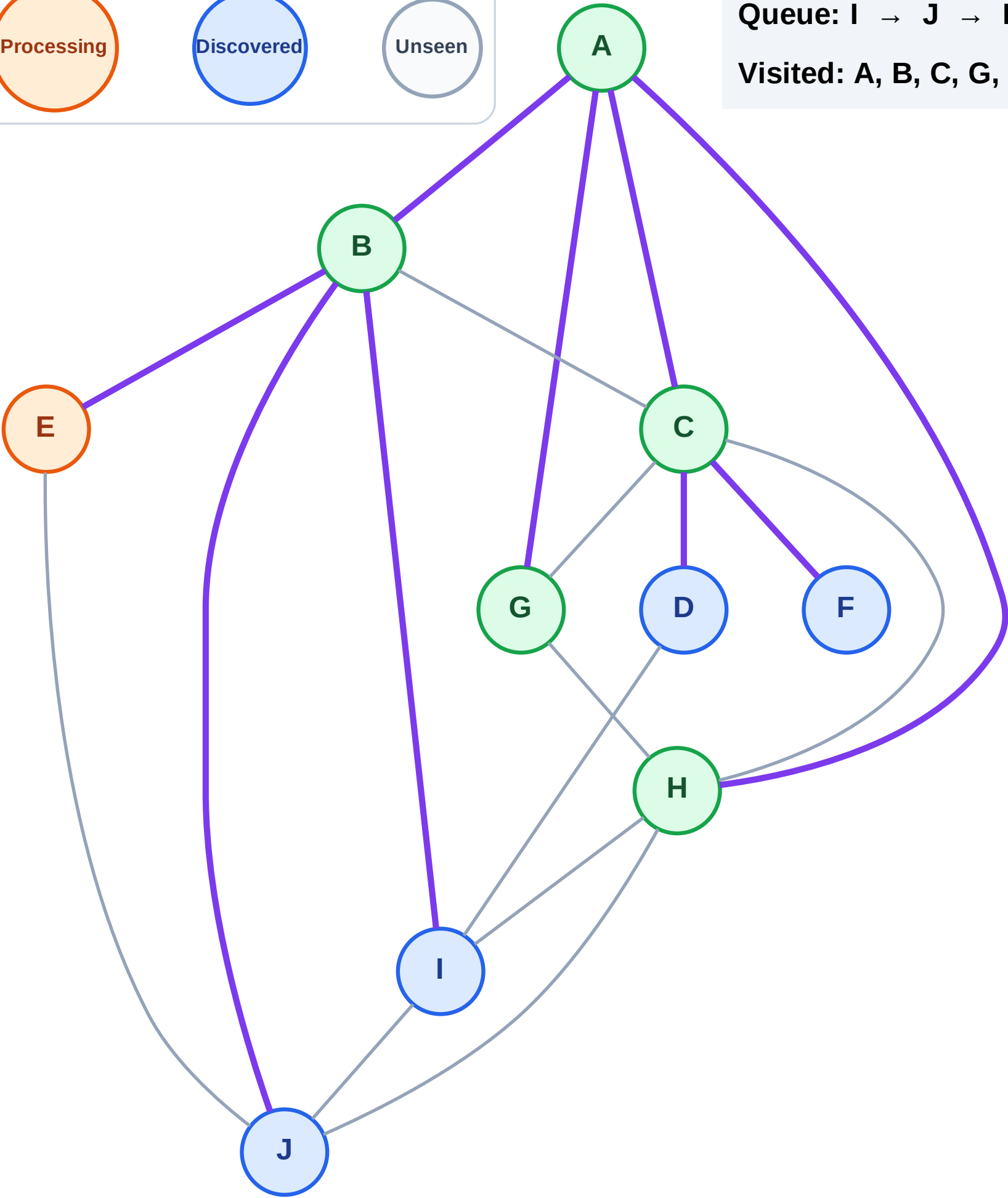
Processing

Discovered

Unseen

Queue: I → J → D → F

Visited: A, B, C, G, H



BFS Traversal

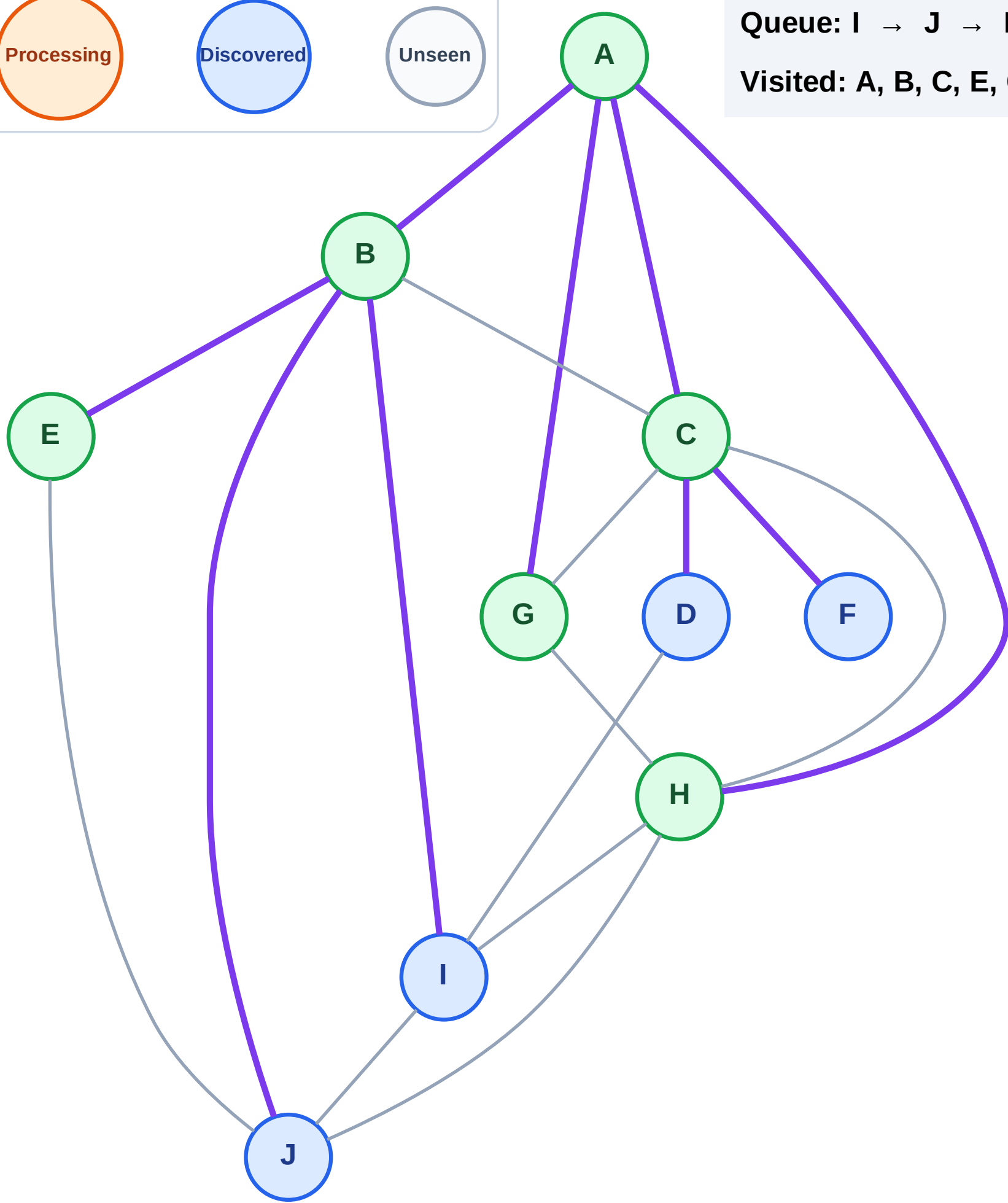
Step 13: No new neighbors from E

Legend



Queue: I → J → D → F

Visited: A, B, C, E, G, H



BFS Traversal

Step 14: Process node I

Legend

Complete

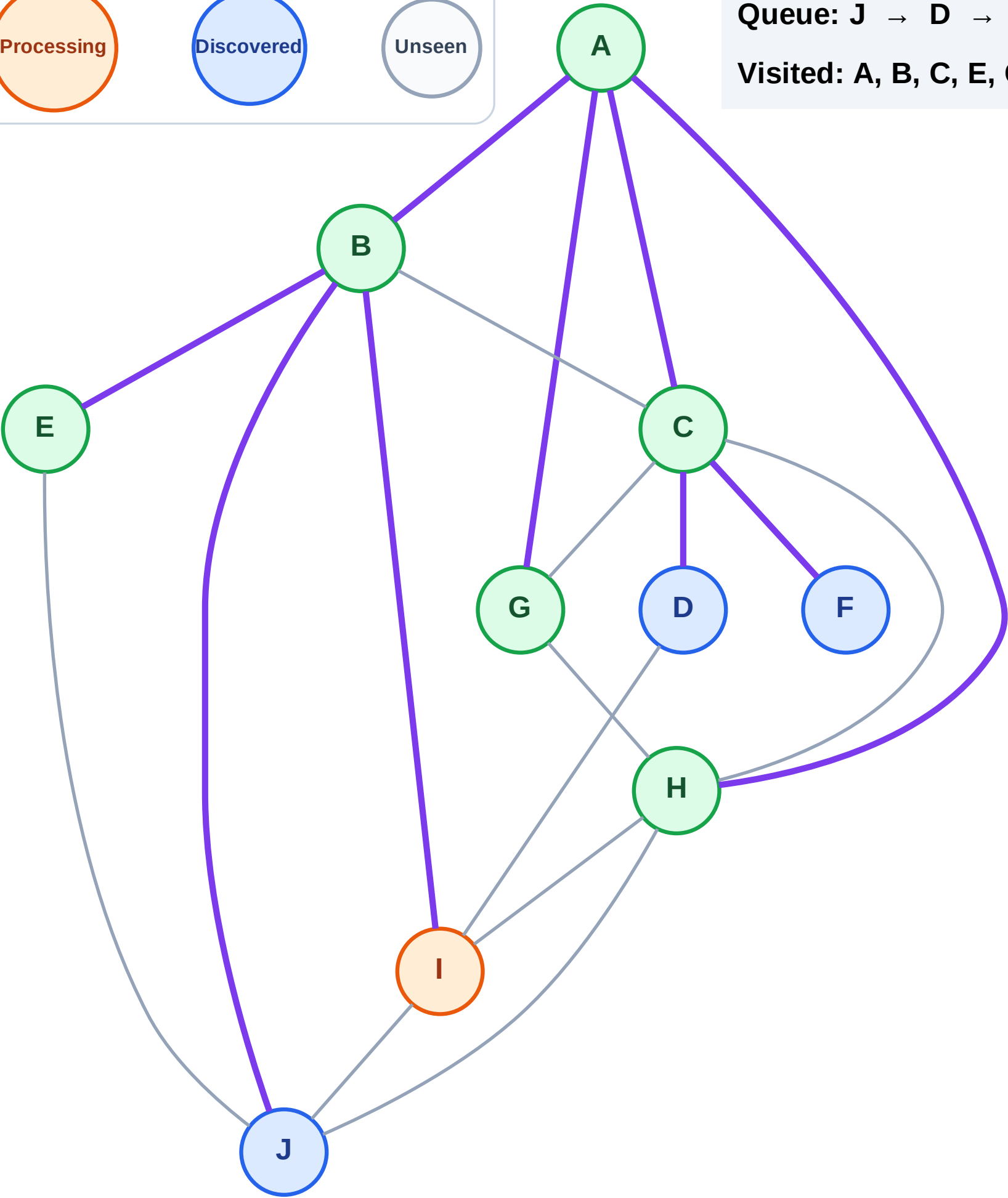
Processing

Discovered

Unseen

Queue: J → D → F

Visited: A, B, C, E, G, H



BFS Traversal

Step 15: No new neighbors from I

Legend

Complete

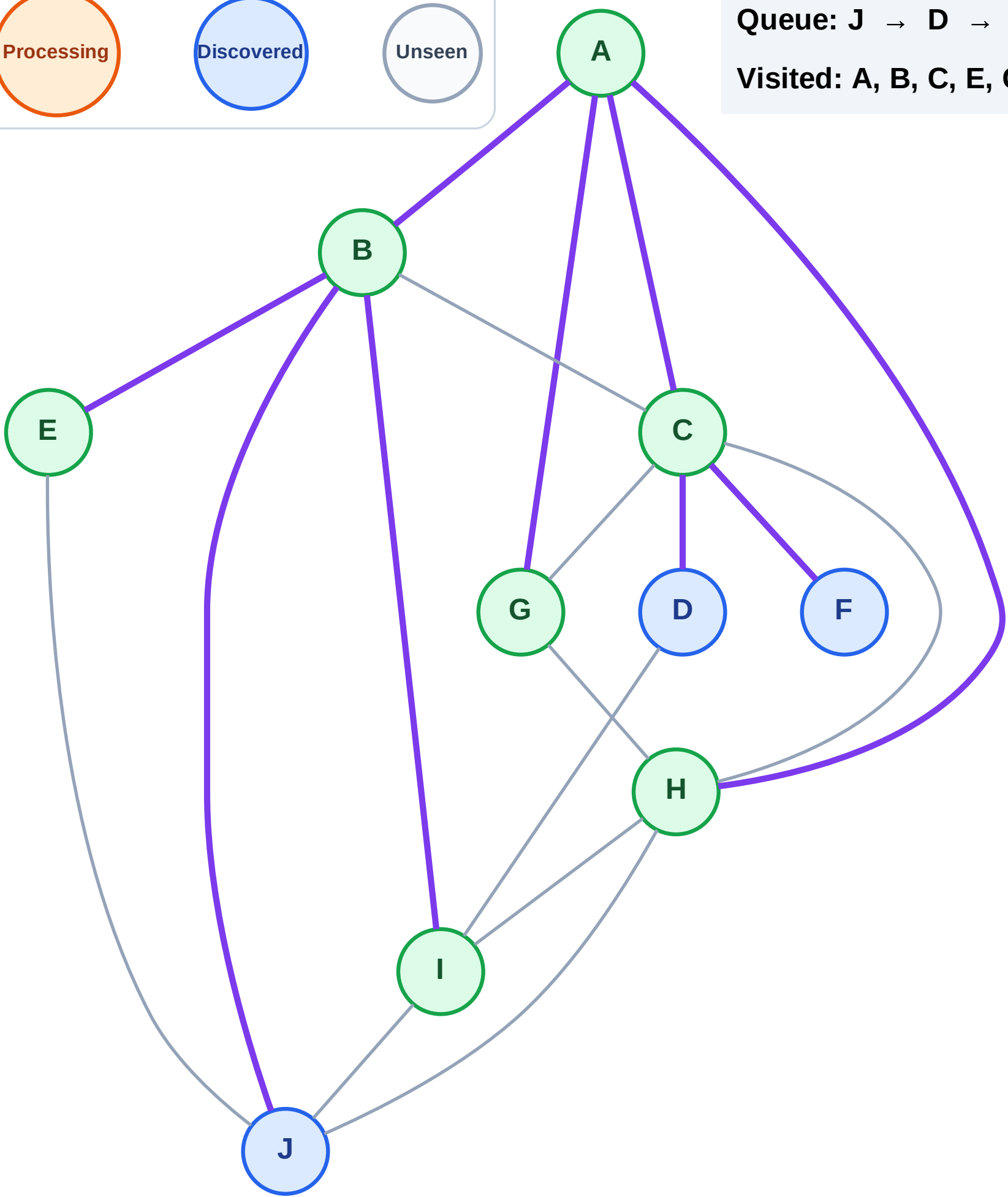
Processing

Discovered

Unseen

Queue: J → D → F

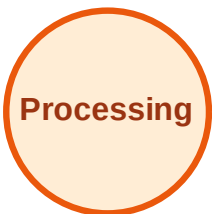
Visited: A, B, C, E, G, H, I



BFS Traversal

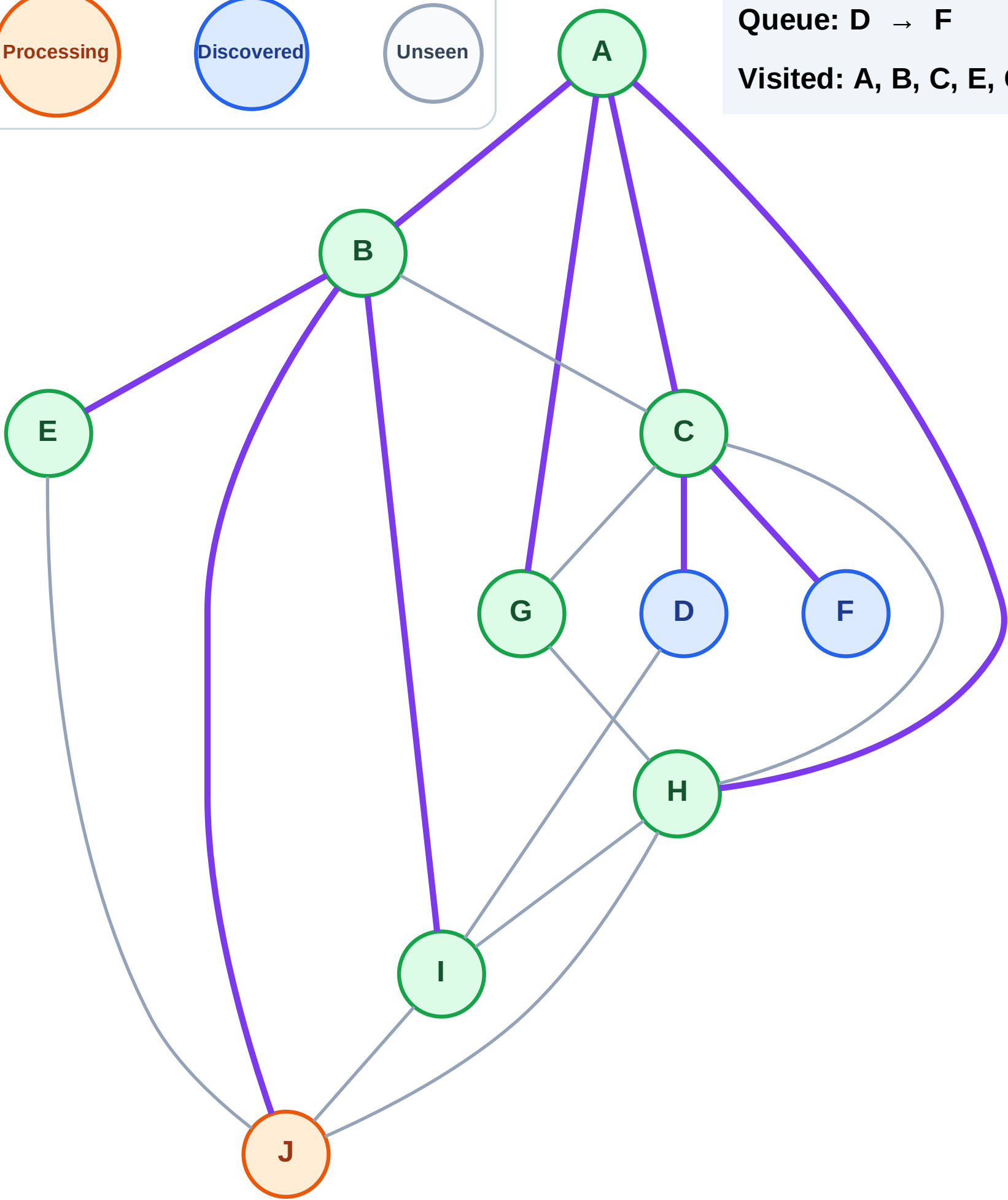
Step 16: Process node J

Legend



Queue: D → F

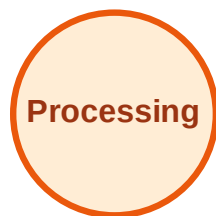
Visited: A, B, C, E, G, H, I



BFS Traversal

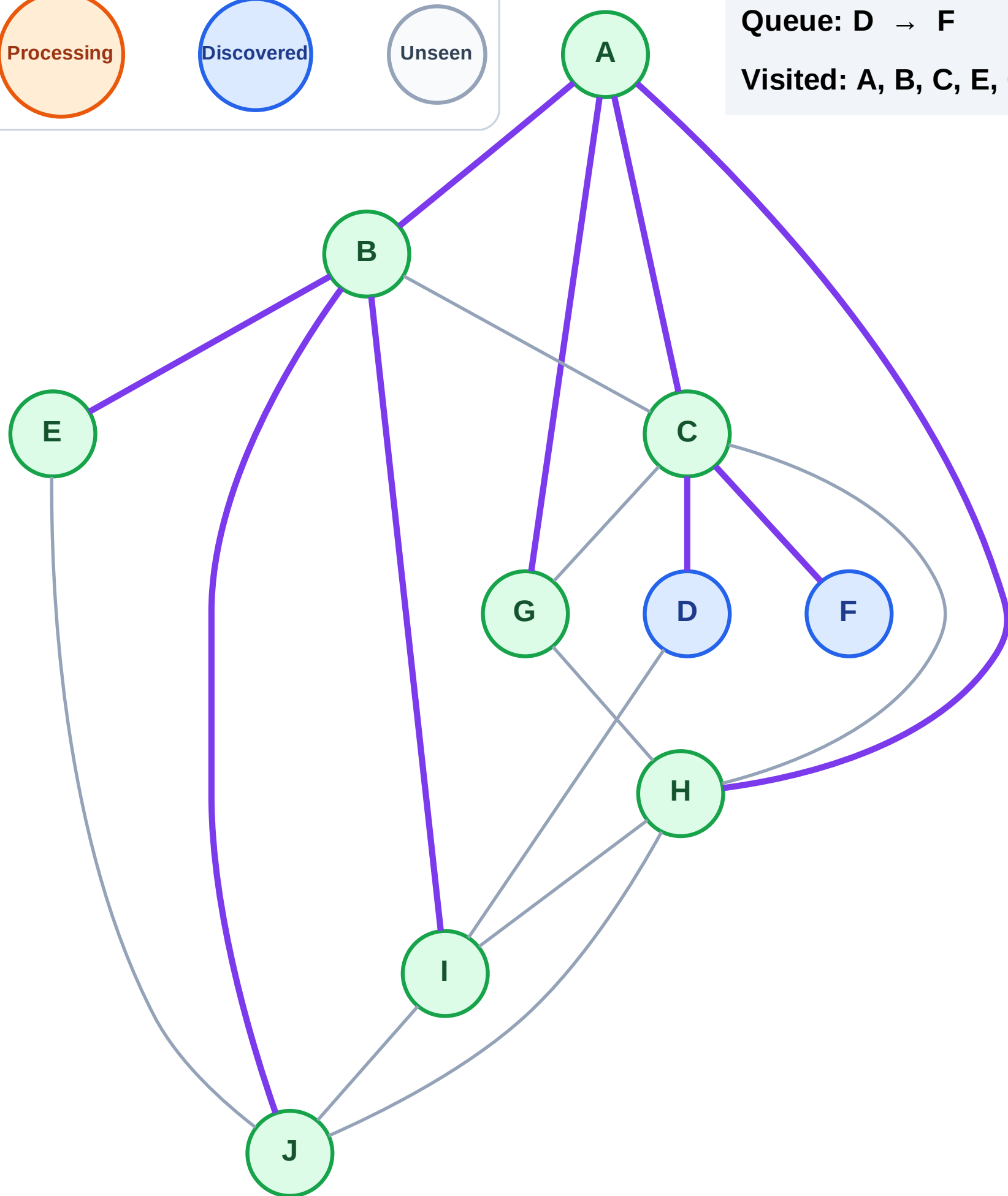
Step 17: No new neighbors from J

Legend



Queue: D → F

Visited: A, B, C, E, G, H, I, J



BFS Traversal

Step 18: Process node D

Legend

Complete

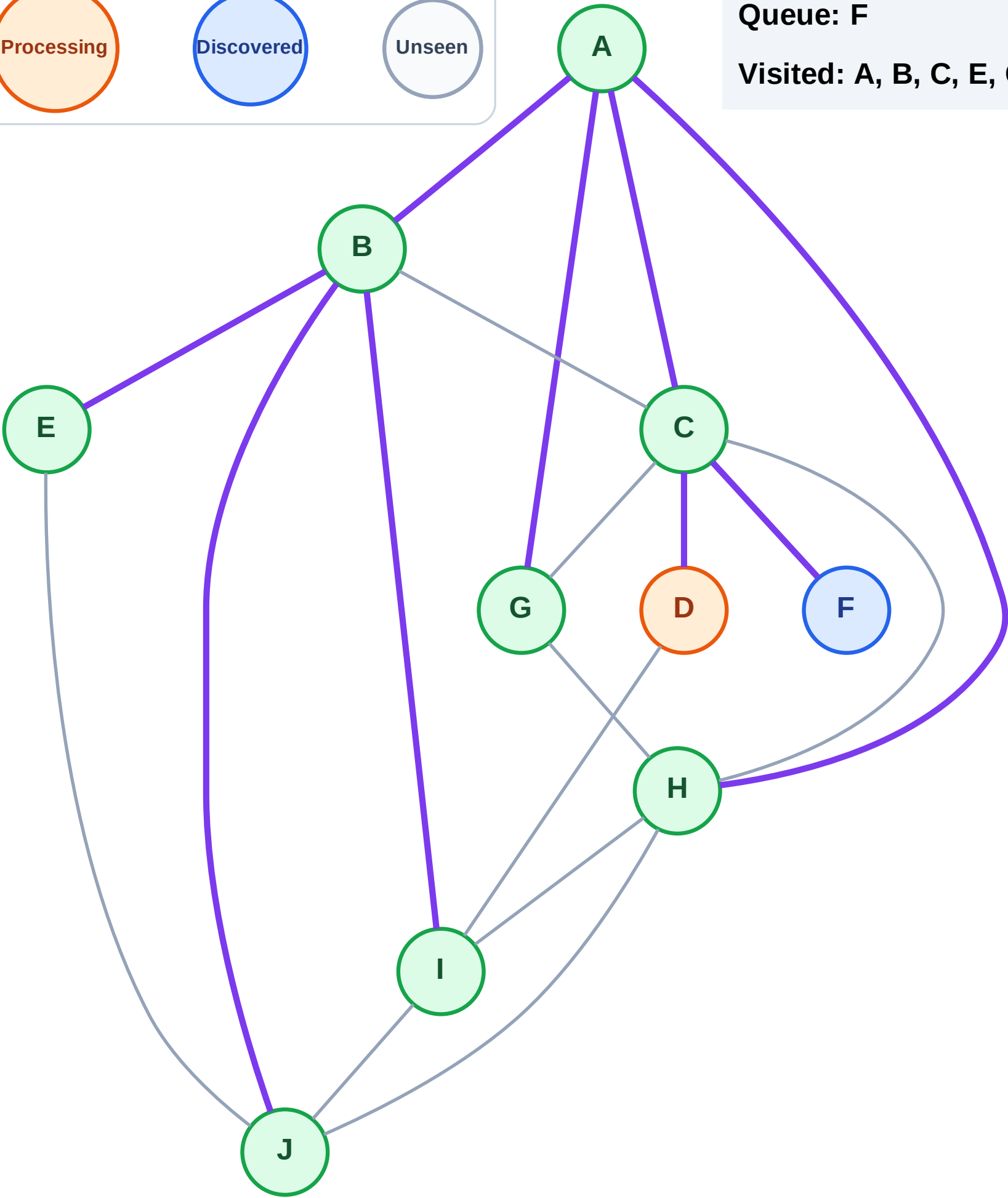
Processing

Discovered

Unseen

Queue: F

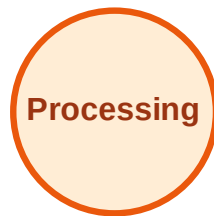
Visited: A, B, C, E, G, H, I, J



BFS Traversal

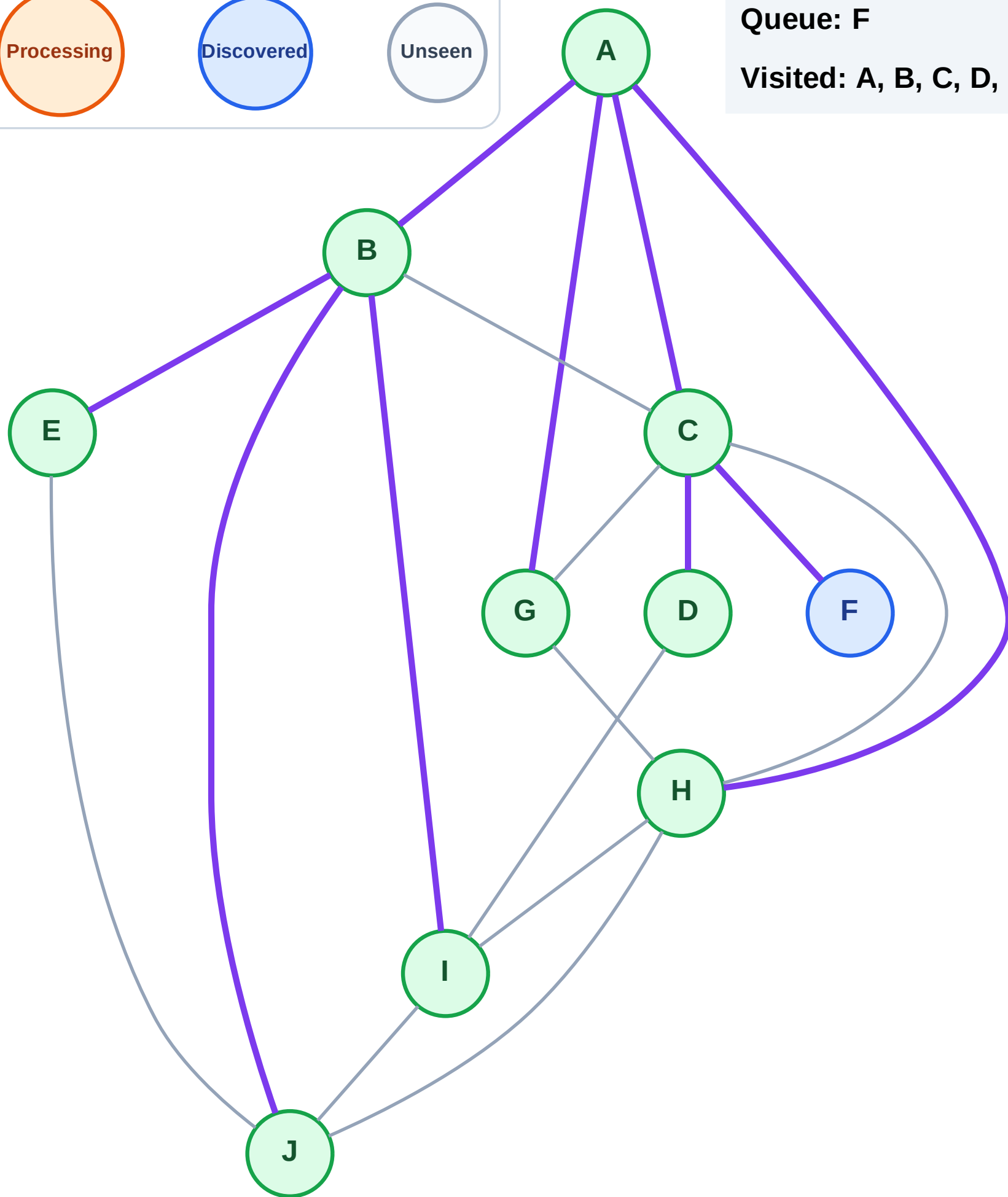
Step 19: No new neighbors from D

Legend



Queue: F

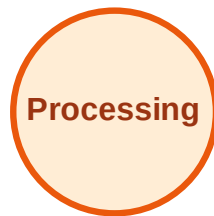
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

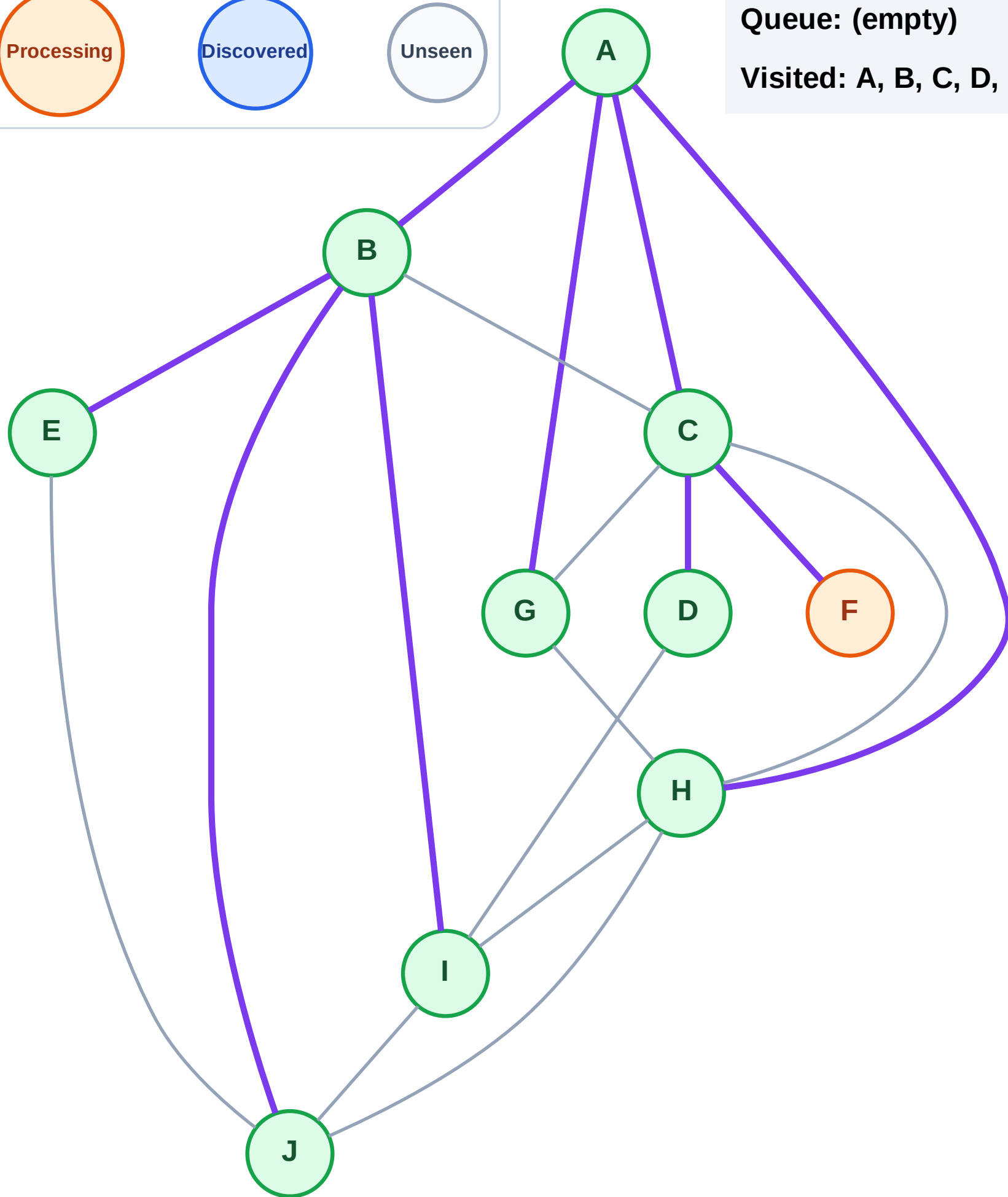
Step 20: Process node F

Legend



Queue: (empty)

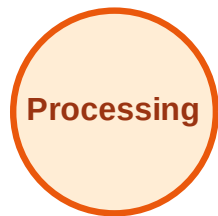
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

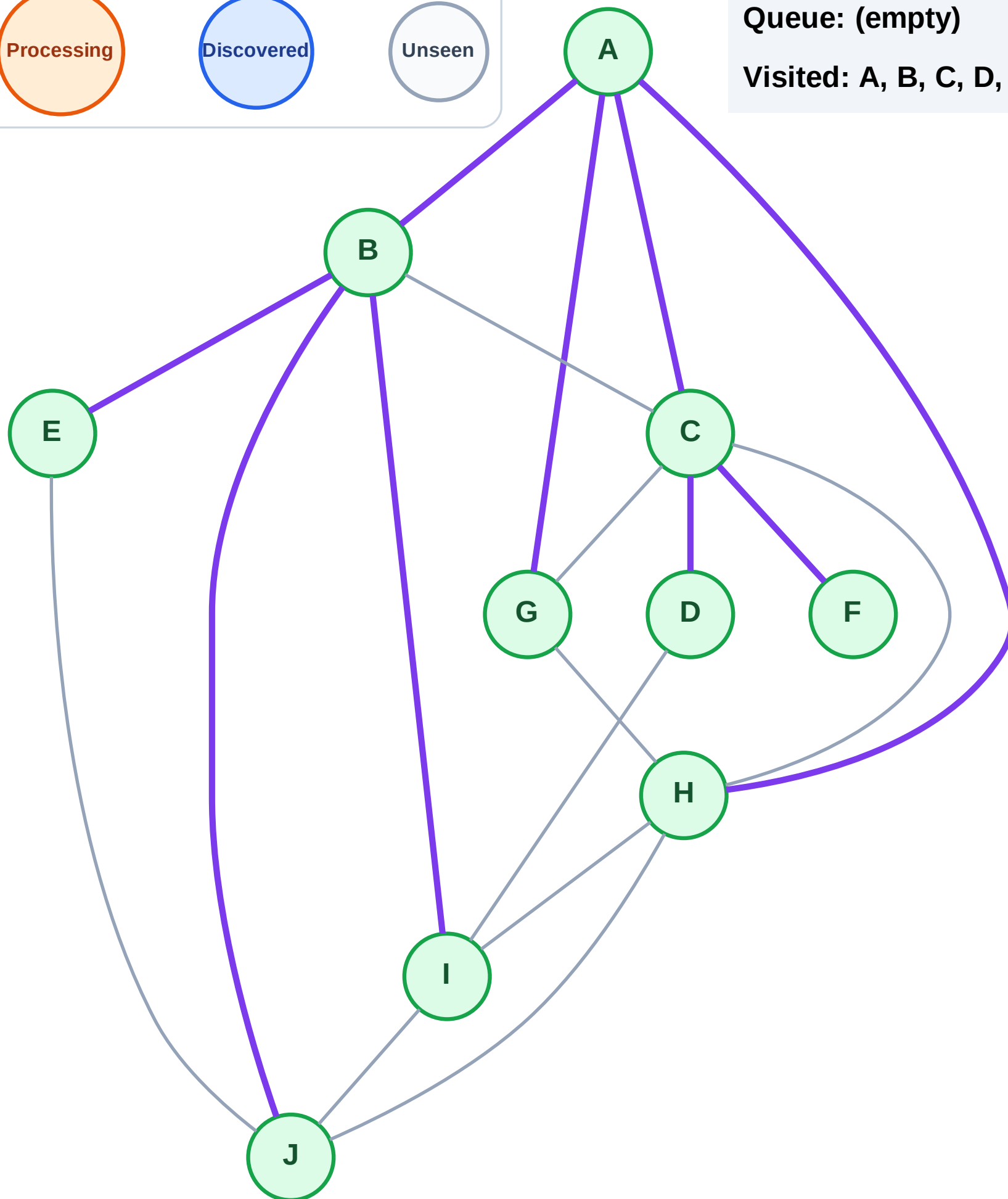
Step 21: No new neighbors from F

Legend



Queue: (empty)

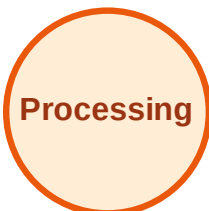
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

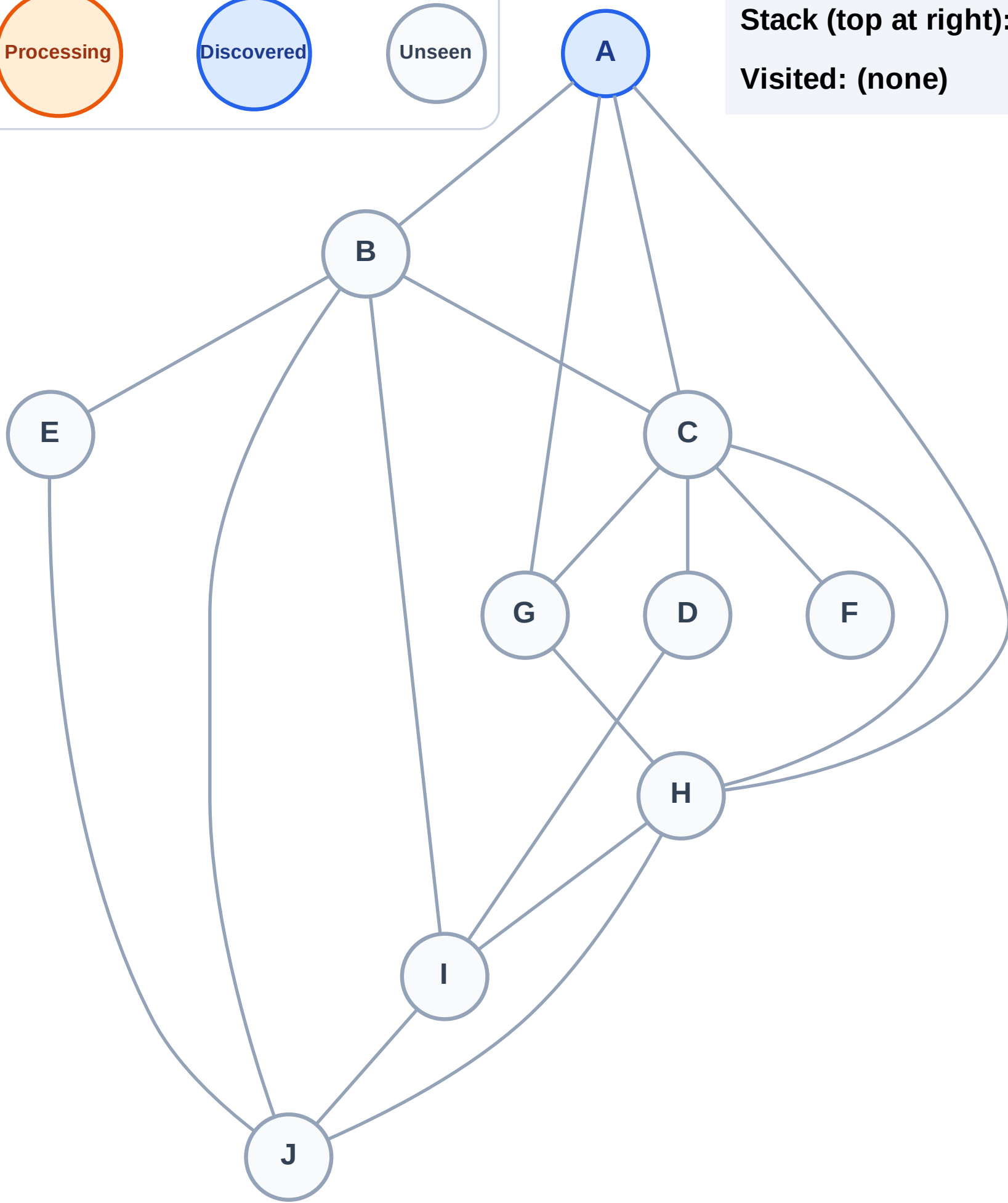
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

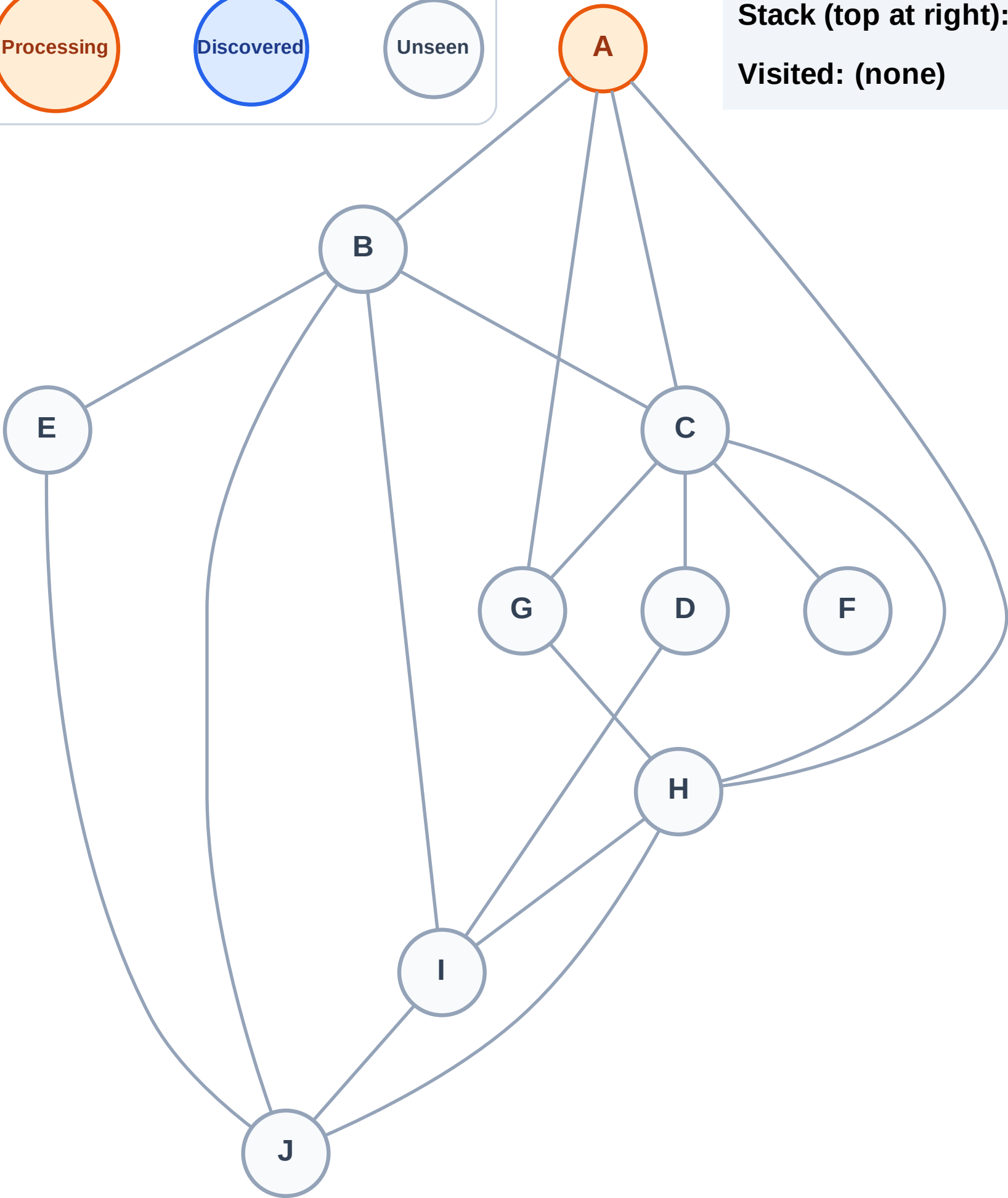
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, G, C, B from A

Legend

Complete

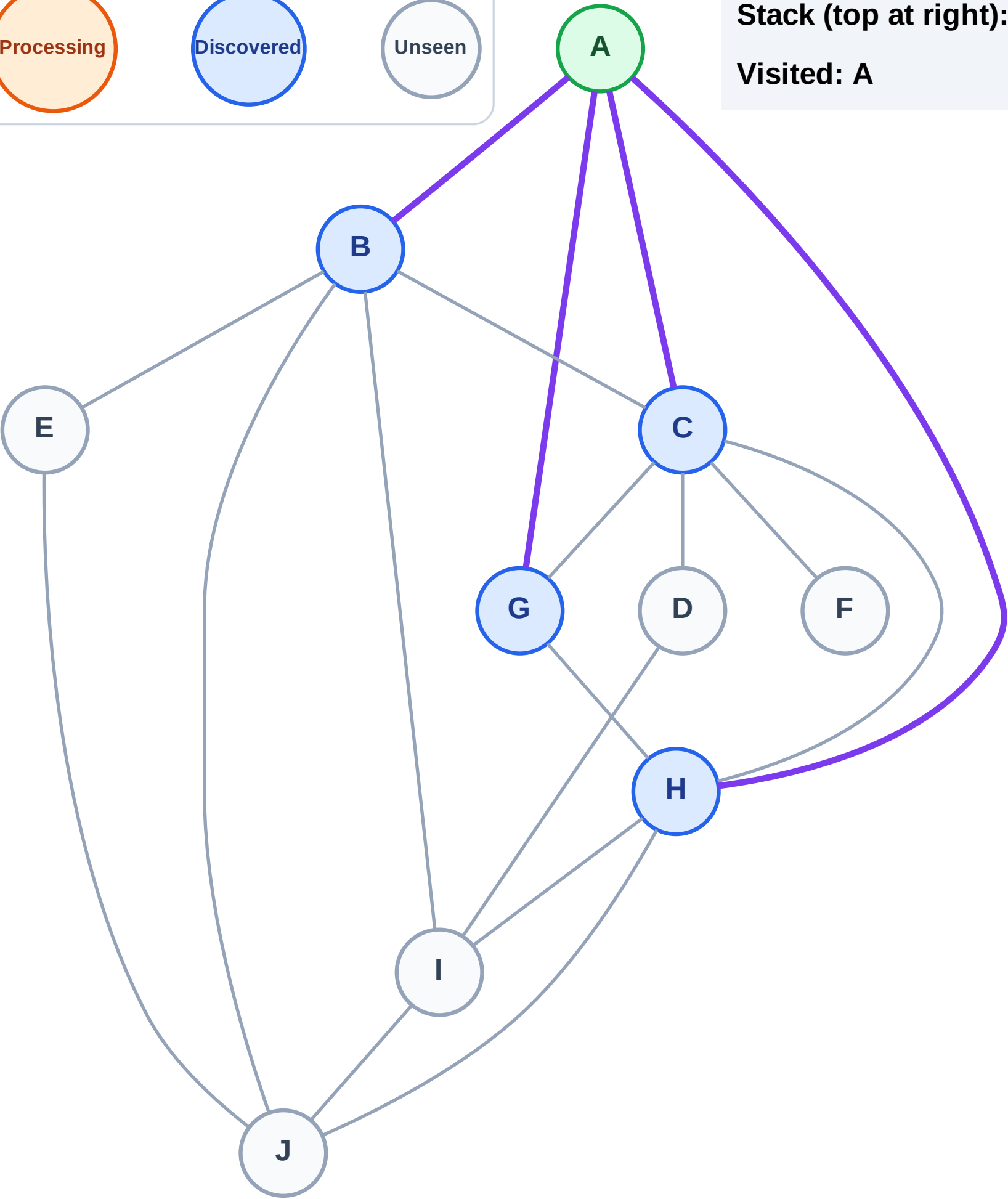
Processing

Discovered

Unseen

Stack (top at right): H | G | C | B

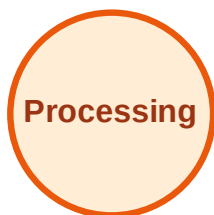
Visited: A



DFS Traversal

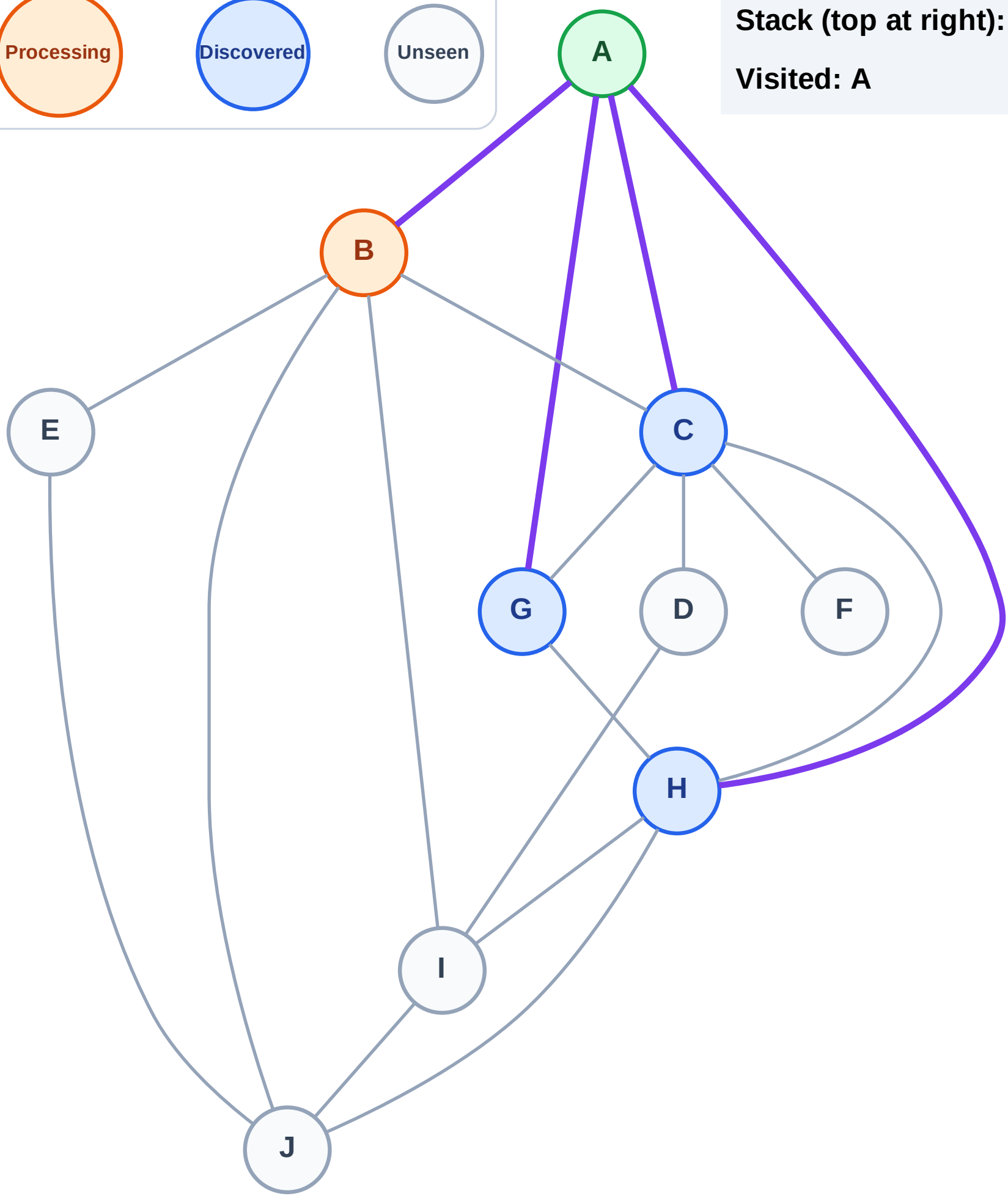
Step 4: Process node B

Legend



Stack (top at right): H | G | C

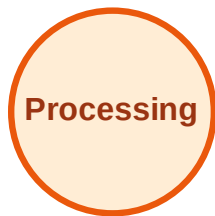
Visited: A



DFS Traversal

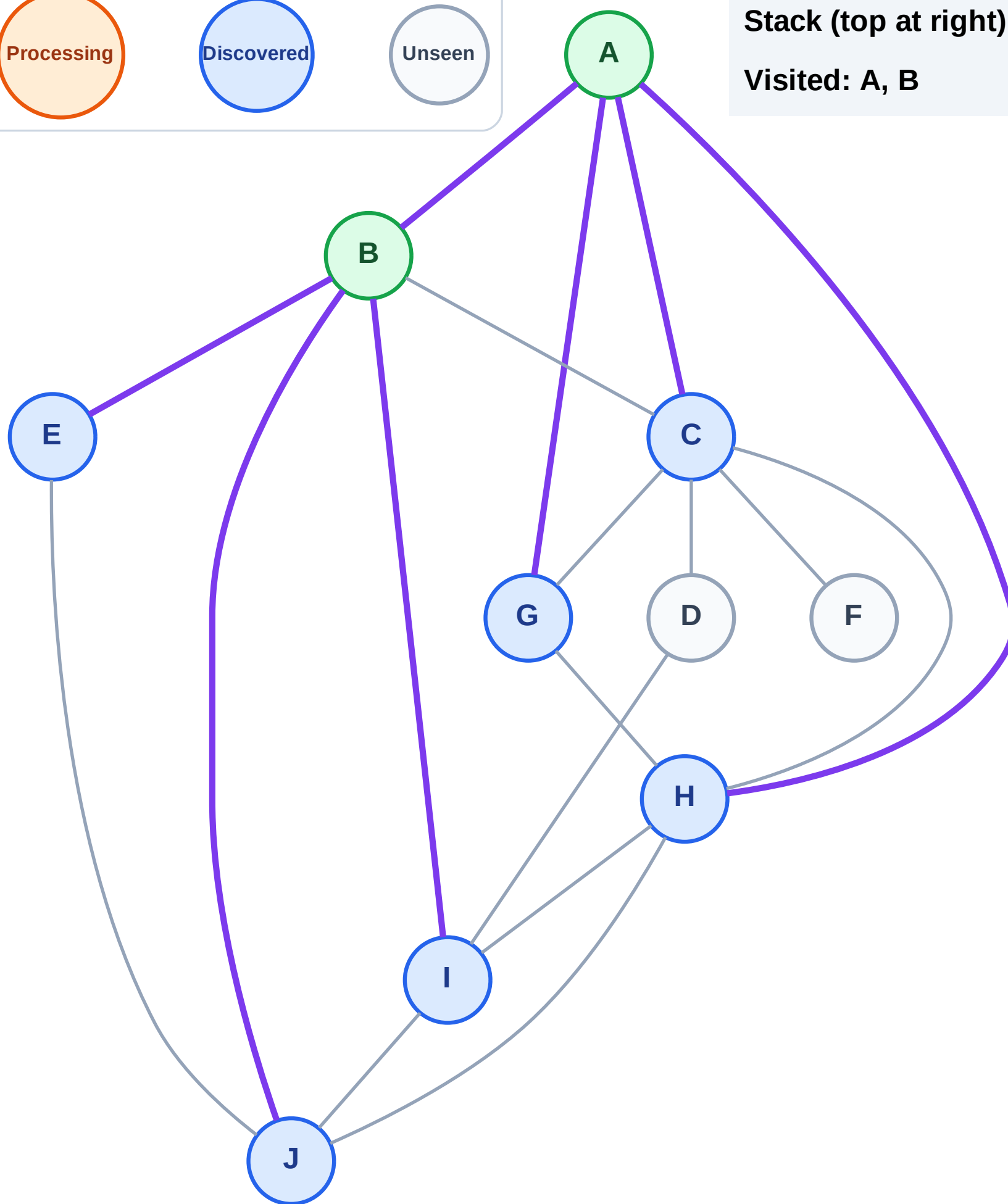
Step 5: Discover J, I, E from B

Legend



Stack (top at right): H | G | C | J | I | E

Visited: A, B



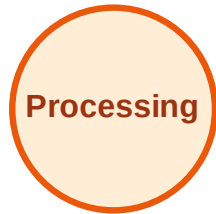
DFS Traversal

Step 6: Process node E

Legend



Complete



Processing



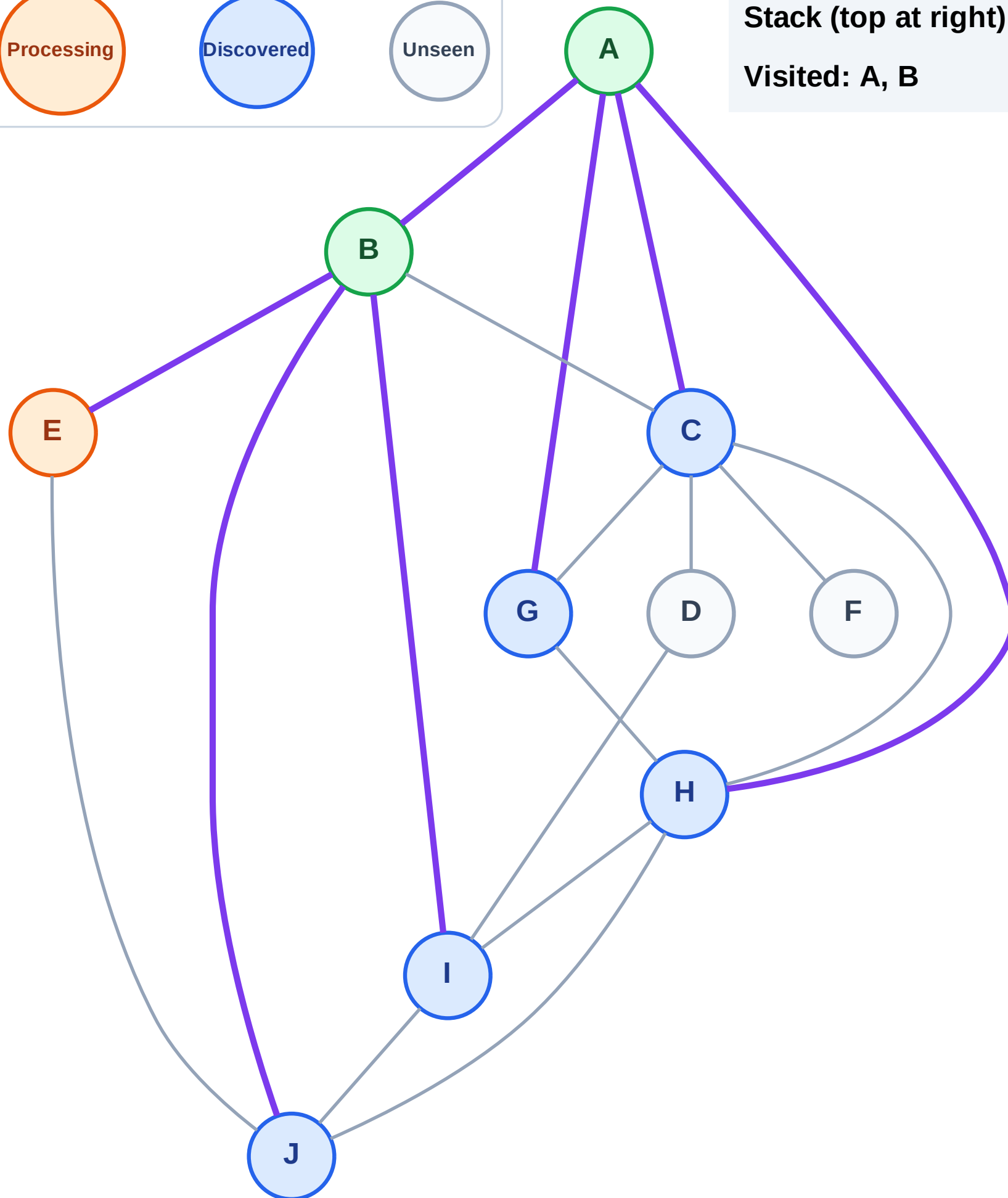
Discovered



Unseen

Stack (top at right): H | G | C | J | I

Visited: A, B



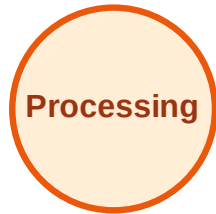
DFS Traversal

Step 7: No new neighbors from E

Legend



Complete



Processing



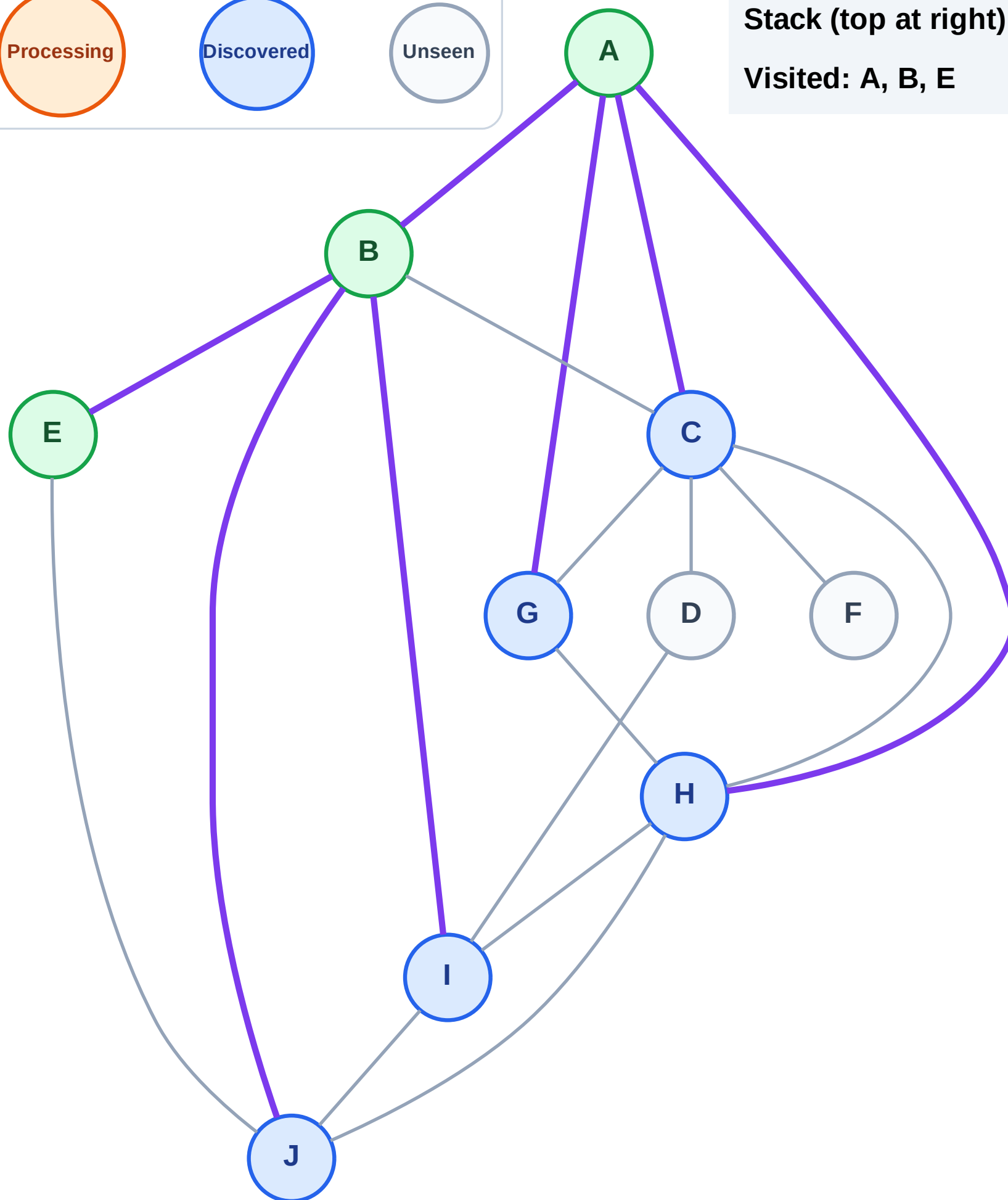
Discovered



Unseen

Stack (top at right): H | G | C | J | I

Visited: A, B, E



DFS Traversal

Step 8: Process node I

Legend

Complete

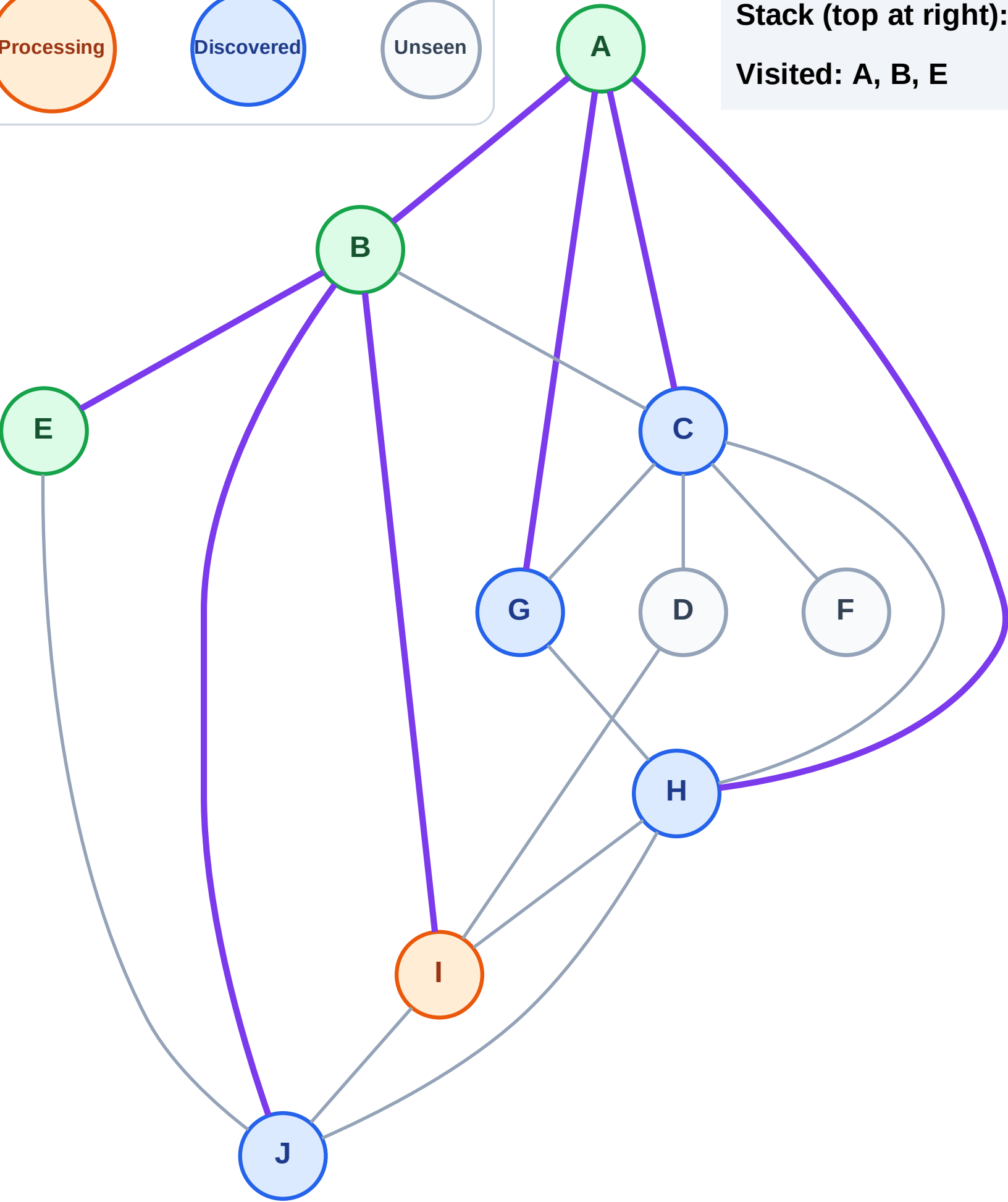
Processing

Discovered

Unseen

Stack (top at right): H | G | C | J

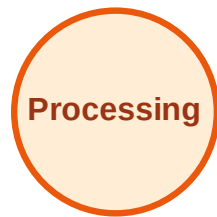
Visited: A, B, E



DFS Traversal

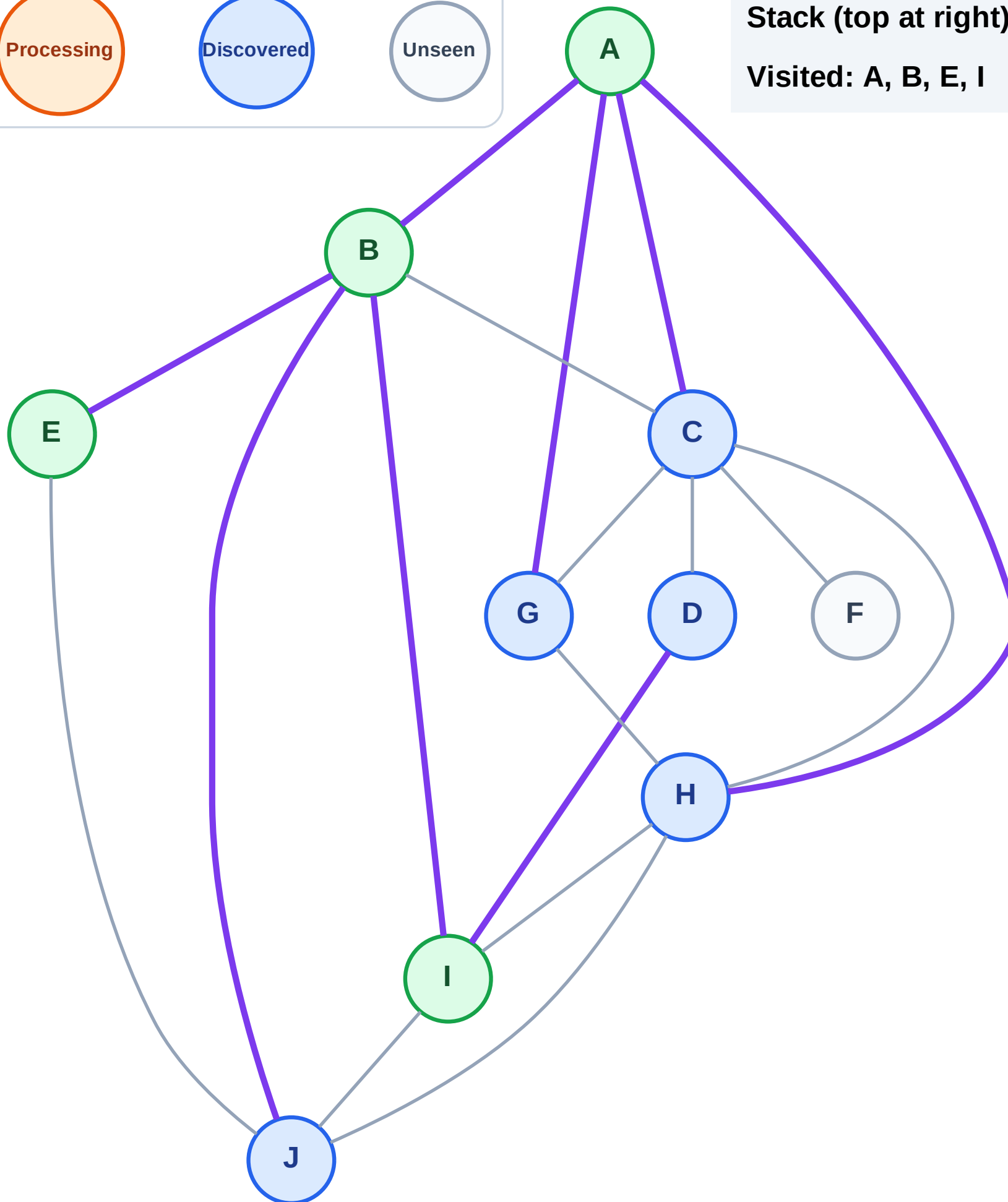
Step 9: Discover D from I

Legend



Stack (top at right): H | G | C | J | D

Visited: A, B, E, I



DFS Traversal

Step 10: Process node D

Legend

Complete

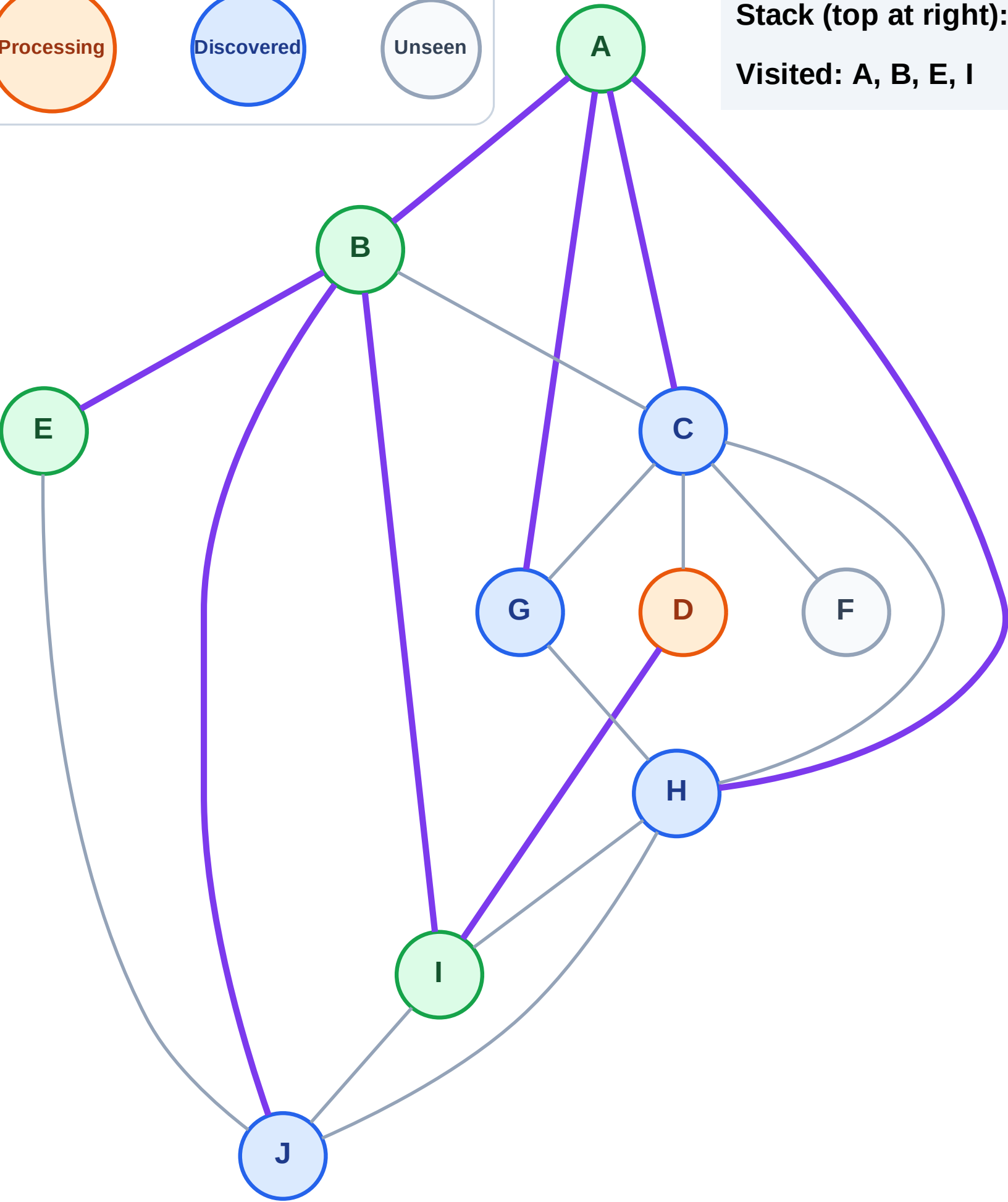
Processing

Discovered

Unseen

Stack (top at right): H | G | C | J

Visited: A, B, E, I



DFS Traversal

Step 11: No new neighbors from D

Legend

Complete

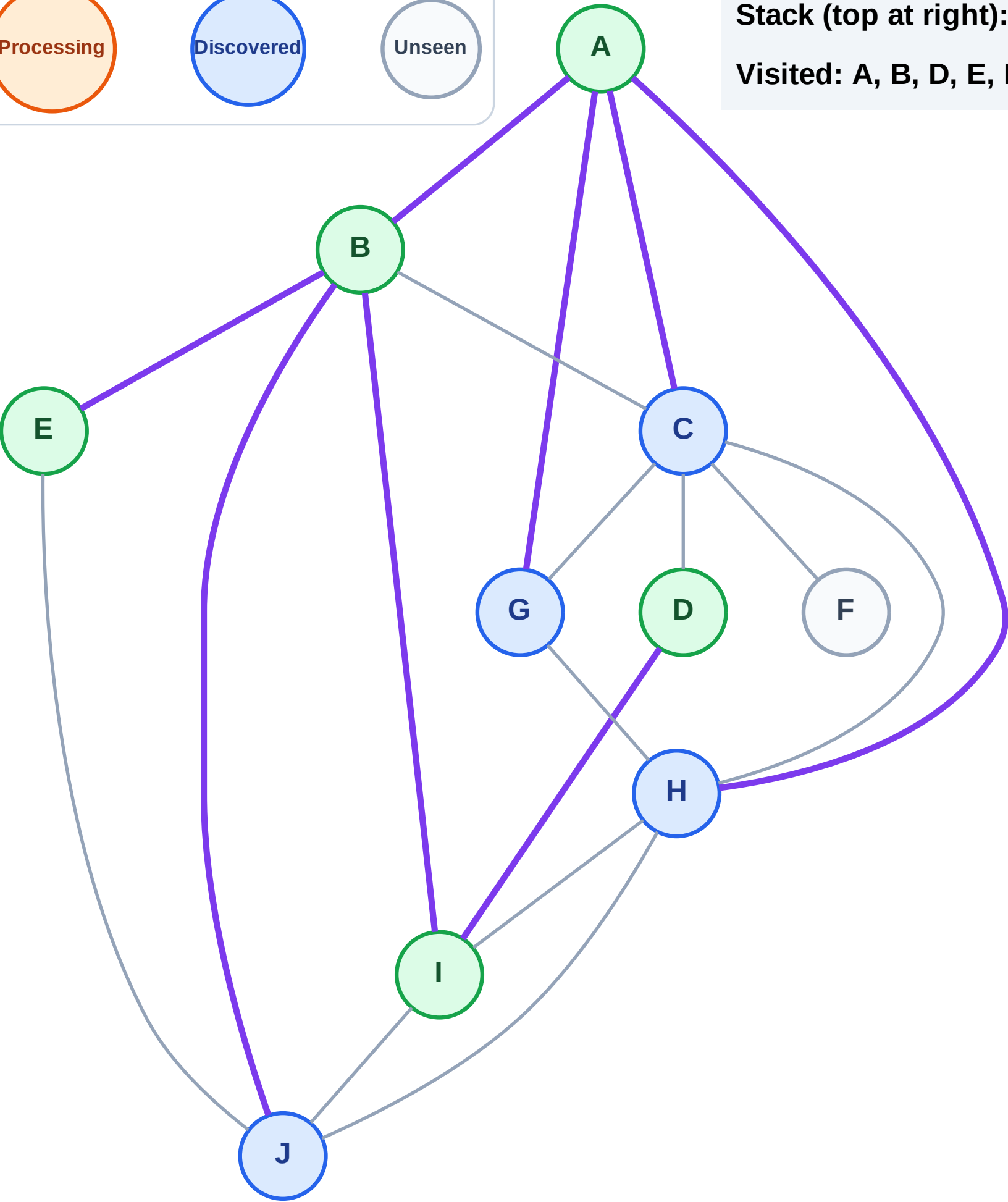
Processing

Discovered

Unseen

Stack (top at right): H | G | C | J

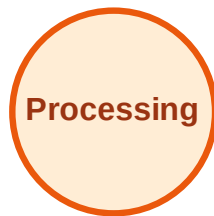
Visited: A, B, D, E, I



DFS Traversal

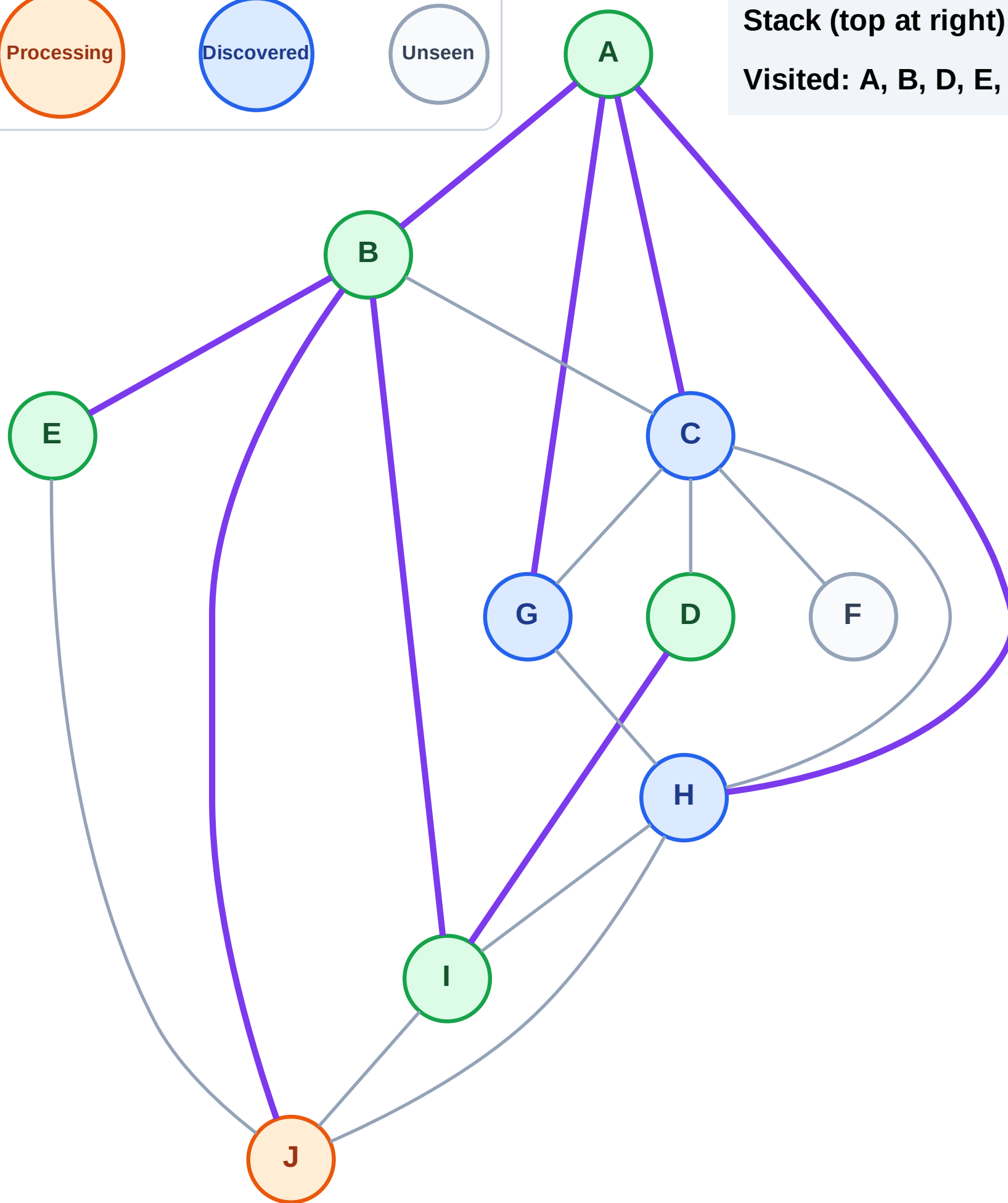
Step 12: Process node J

Legend



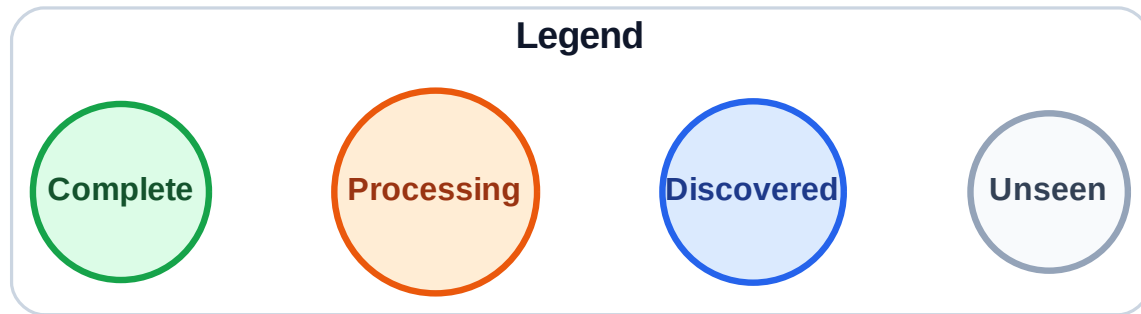
Stack (top at right): H | G | C

Visited: A, B, D, E, I



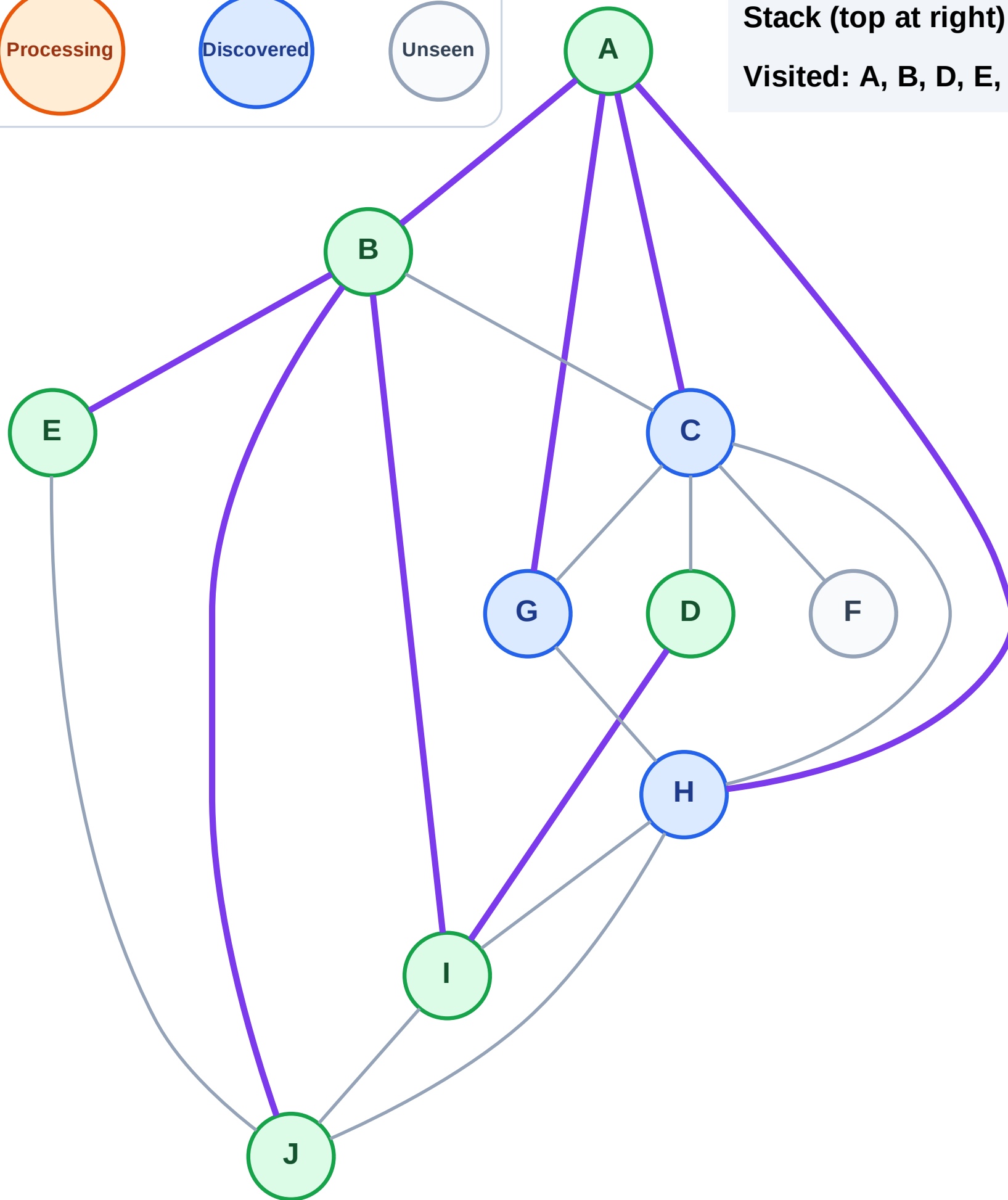
Step 13: No new neighbors from J

Step 13: No new neighbors from J



Stack (top at right): H | G | C

Visited: A, B, D, E, I, J



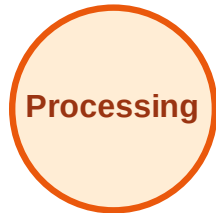
DFS Traversal

Step 14: Process node C

Legend



Complete



Processing



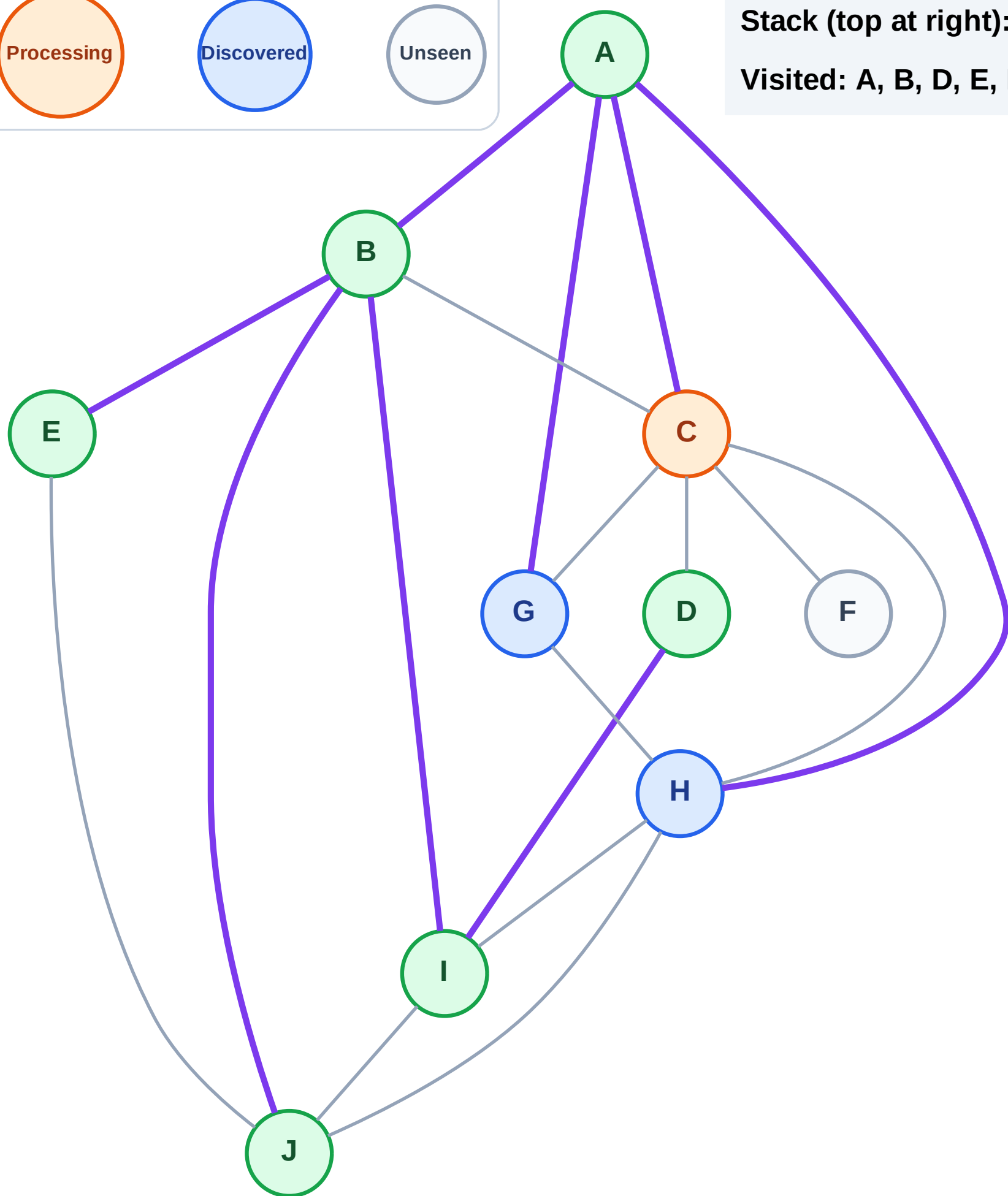
Discovered



Unseen

Stack (top at right): H | G

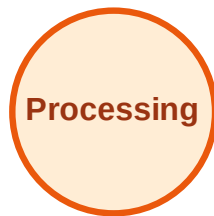
Visited: A, B, D, E, I, J



DFS Traversal

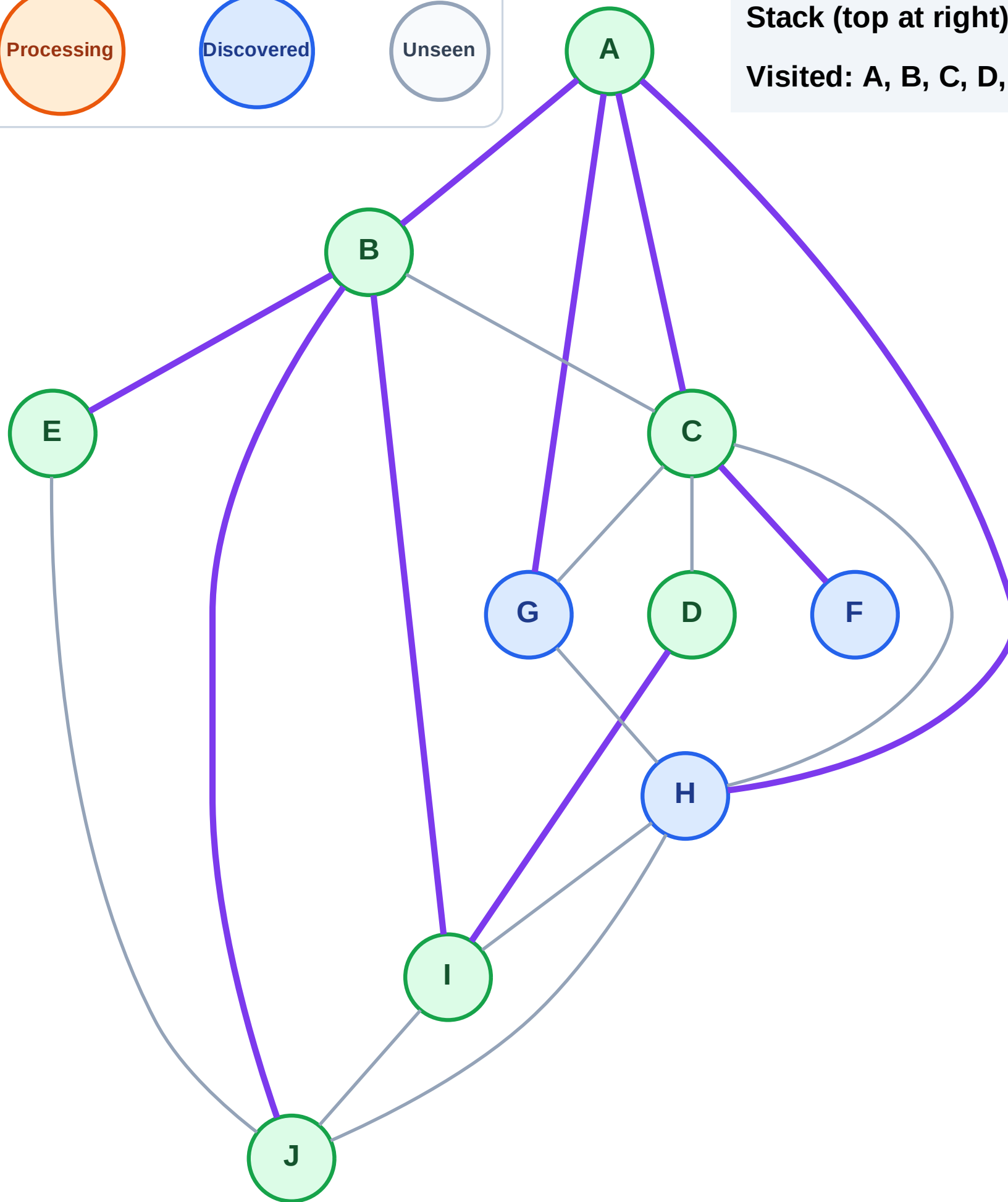
Step 15: Discover F from C

Legend



Stack (top at right): H | G | F

Visited: A, B, C, D, E, I, J



DFS Traversal

Step 16: Process node F

Legend

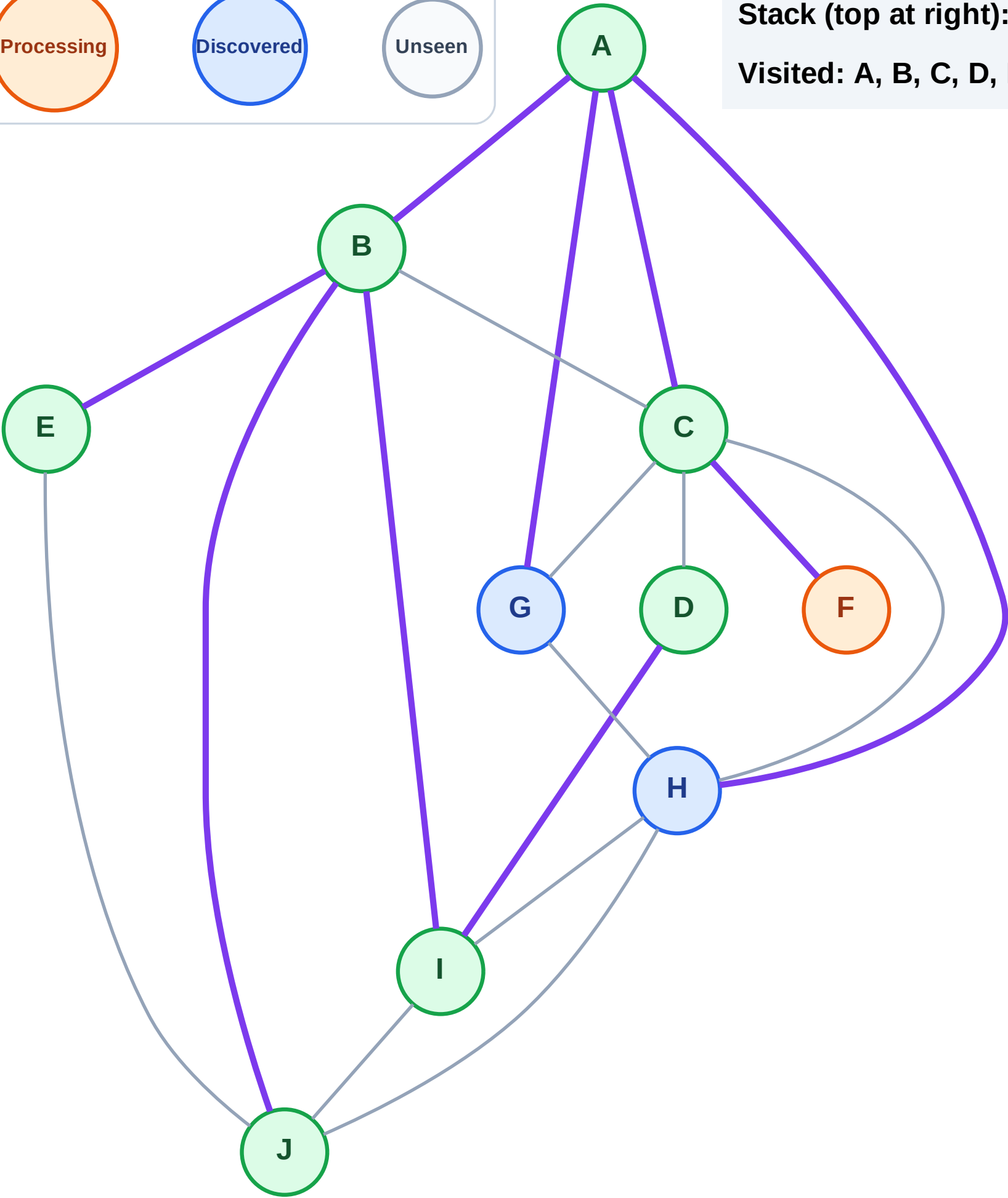
Complete

Processing

Discovered

Unseen

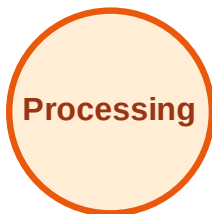
Stack (top at right): H | G
Visited: A, B, C, D, E, I, J



DFS Traversal

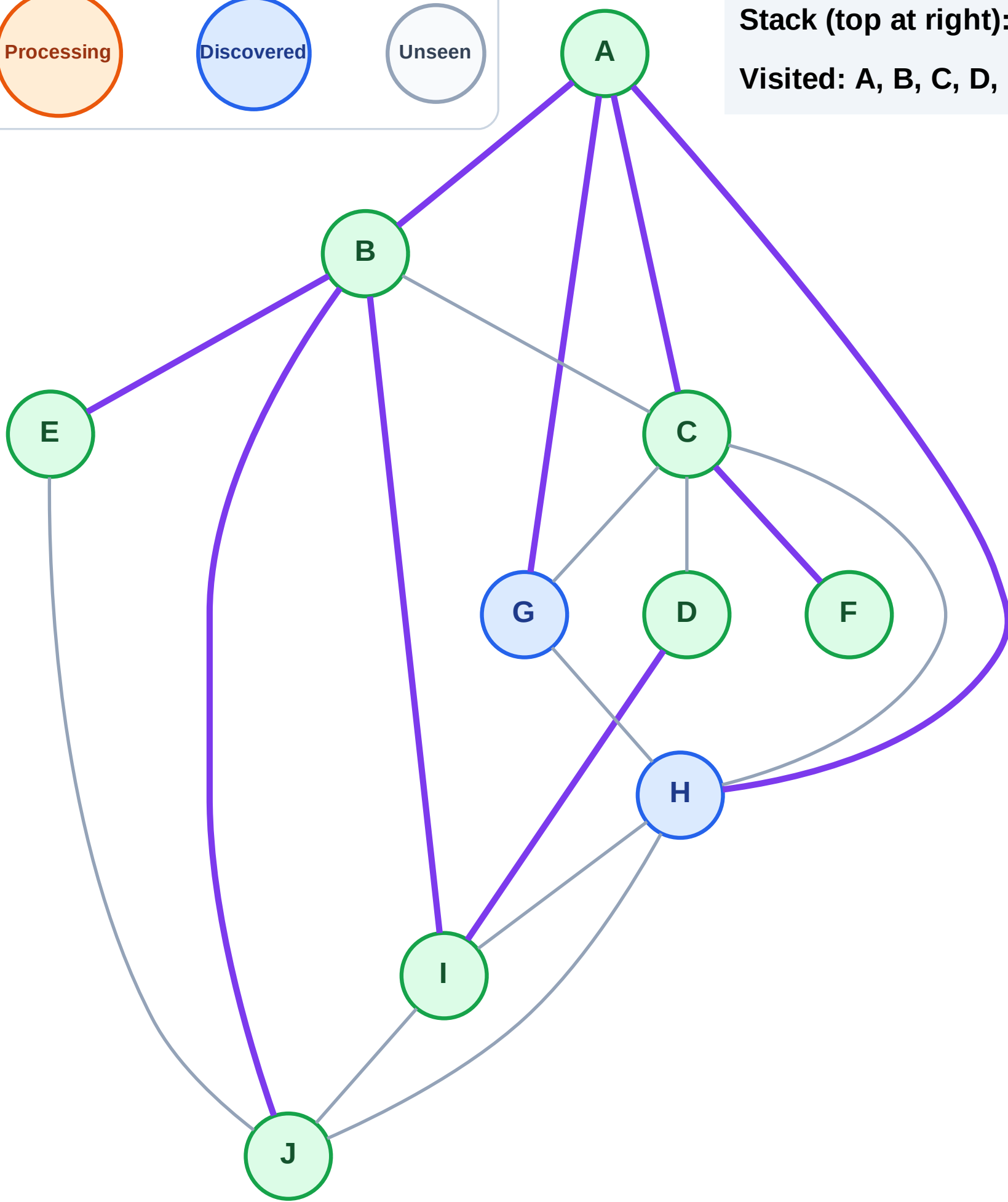
Step 17: No new neighbors from F

Legend



Stack (top at right): H | G

Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 18: Process node G

Legend

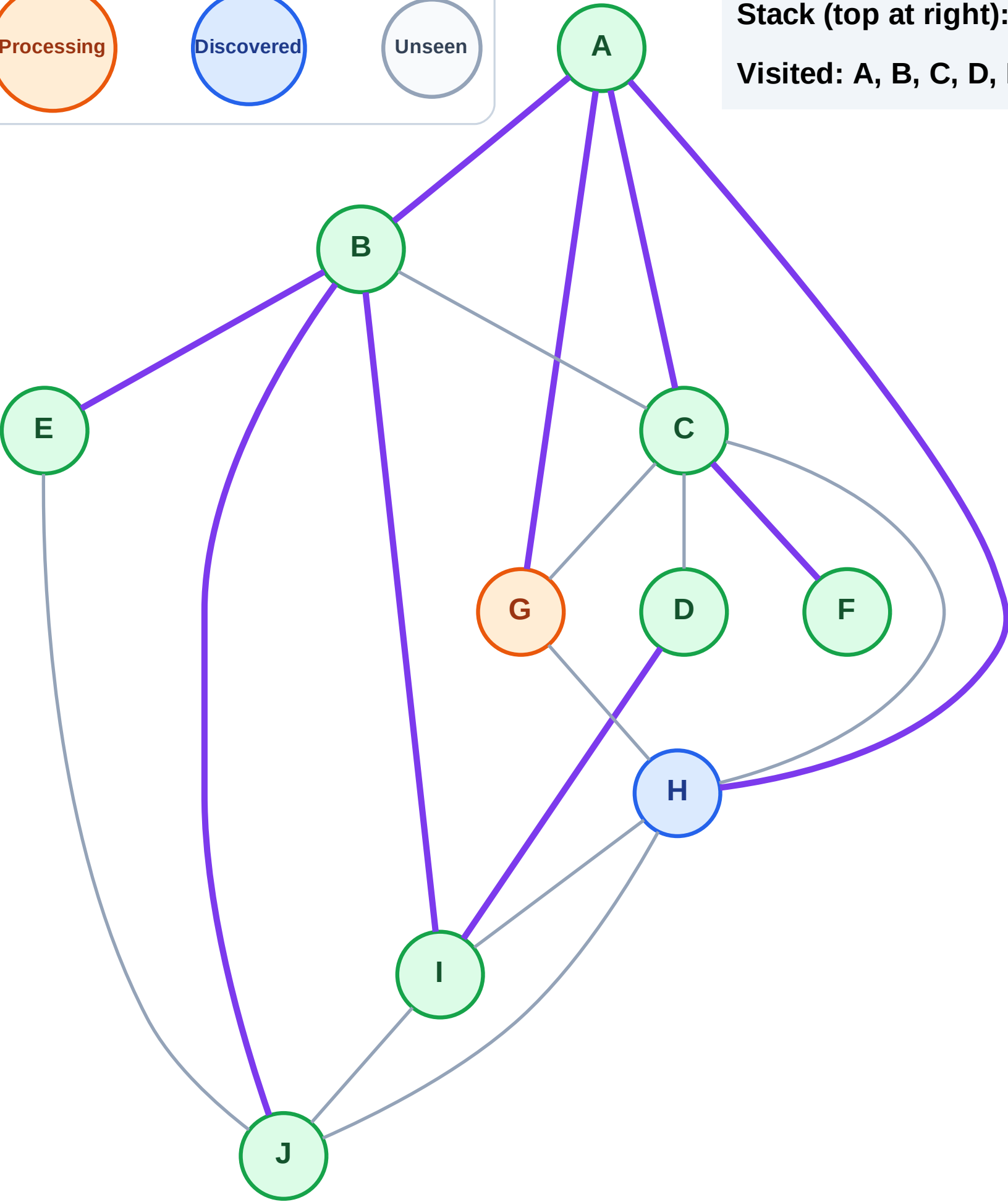
Complete

Processing

Discovered

Unseen

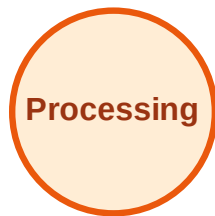
Stack (top at right): H
Visited: A, B, C, D, E, F, I, J



DFS Traversal

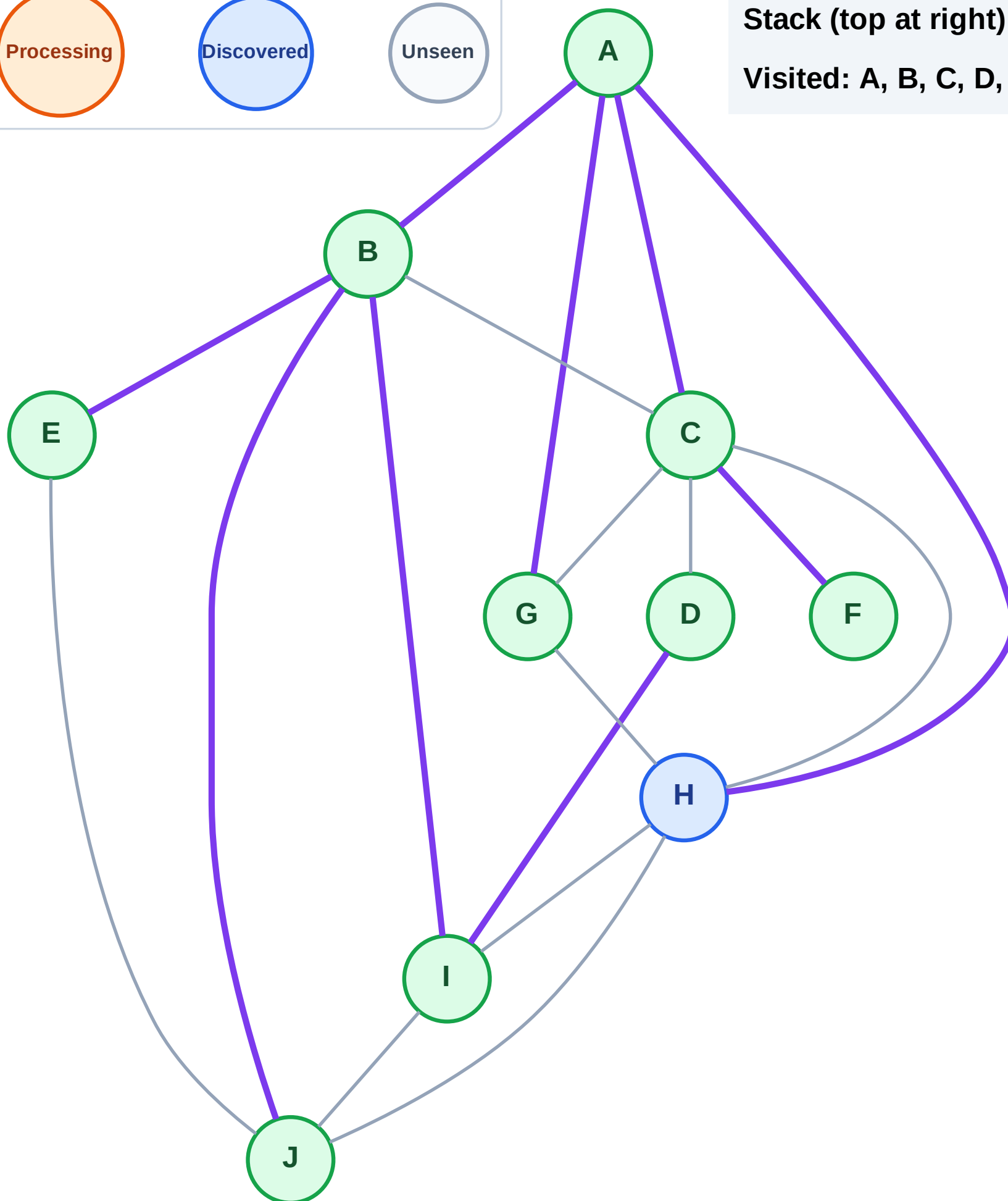
Step 19: No new neighbors from G

Legend



Stack (top at right): H

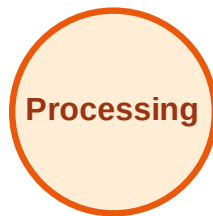
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

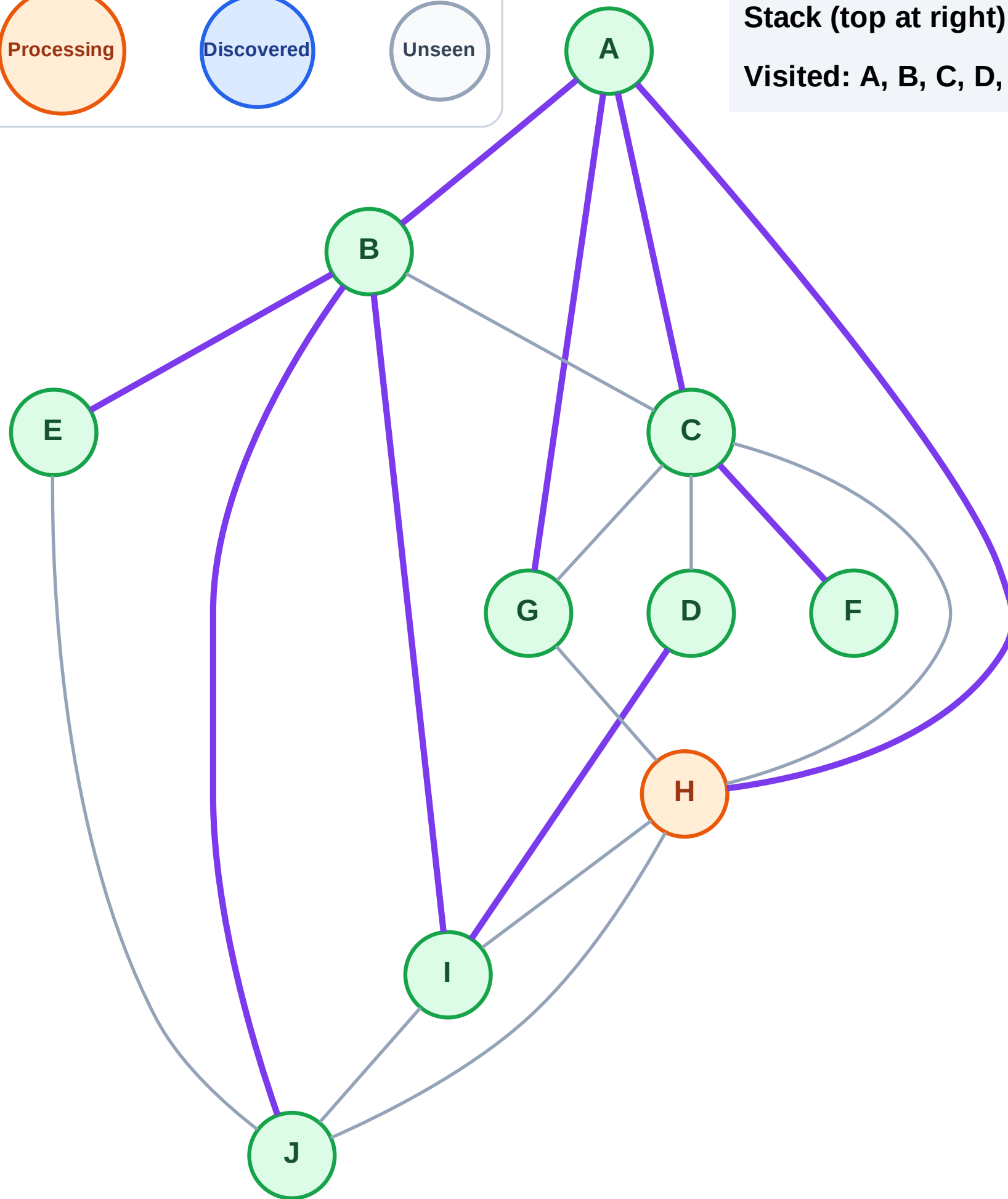
Step 20: Process node H

Legend



Stack (top at right): (empty)

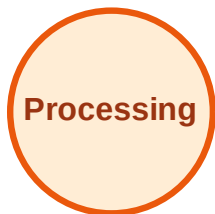
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

